Figure 3 supplementary methods: linear response and the neutral mode

Every deterministic condition starts from its exact Figure 4 basal checkpoint. Uniform mean input is applied without noise and allowed to relax for 16,000 chemical time units or 1,600 Spring time units. A half-contrast edge must be unique, contrast at least 0.05, maximum native RHS no greater than 1e-6, and final 10% preparation change no greater than 1e-4 of the maximum state scale. These gates define a stationary polarized response; failure is recorded rather than replaced by another preparation.

The Gaussian pulse is centered at the interpolated rising or falling half-contrast crossing; its spatial standard deviation equals the declared noise covariance length ell (1.5 chemical, 0.15 Spring), and its duration is tau (50 chemical, 5 Spring). Peak a is 1e-4 of the unchanged native reference: 1e-5 Wave-pinning, 1e-4 Holmes, 4e-4 Spring, 6.4e-4 Goryachev. Each edge receives independent restarts at a/2, a and 2a. The other edge is tracked as well. A no-pulse continuation is subtracted at matching times. Spring edge locations use material fractions times mean-input circumference to remove uniform expansion.

A translated polarized profile is a neutral state. A brief pulse can therefore leave a persistent positional offset after the transient has relaxed: displacement does not have to return to zero. The plotted response is displacement divided by pulse amplitude. The inset is the time integral of that displacement through the explicitly finite 20-tau window, divided by the pulse dose a*tau. It is not an infinite-time integral. Terminal displacement divided by a*tau is the zero-frequency velocity gain.

A single spatial pulse does not identify susceptibility to every location on a ring. The additional spatial response shape is calculated from the unchanged native Jacobian at that same stationary mean-input state. Each edge kernel is normalized to unit gain under that exact pulse shape and multiplied by its measured terminal displacement per pulse dose. The unscaled native gain is separately validated against the measurement. No measured diffusion is used in this calculation. Both edges have the same zero-frequency translation susceptibility; transient shape/width modes need not be the same. The main prediction deliberately applies the requested independent-edge formula to those marginal edge diffusivities.

Linear-response definitions and exact imposed covariance

$$\delta\dot{\mathbf{y}} = J\,\delta\mathbf{y} + B\,\delta\mathbf{u}, \quad J\mathbf{q} = 0, \quad J^{\mathsf{T}}\mathbf{p} = 0, \quad \mathbf{p}^{\mathsf{T}}\mathbf{q} = 1. \tag{1}$$

$$\dot{s} = \sum_j g_j\,\delta u_j, \qquad g_j = (\mathbf{p}^{\mathsf{T}}B)_j. \tag{2}$$

$$K_u(j,k;t,t+v) = \mu^2\{\exp[\sigma^2 C_x(j-k)C_t(t,t+v)] - 1\}. \tag{3}$$

$$D_e = \sum_{j,k} g_j g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\,\mathrm{d}v. \tag{4}$$

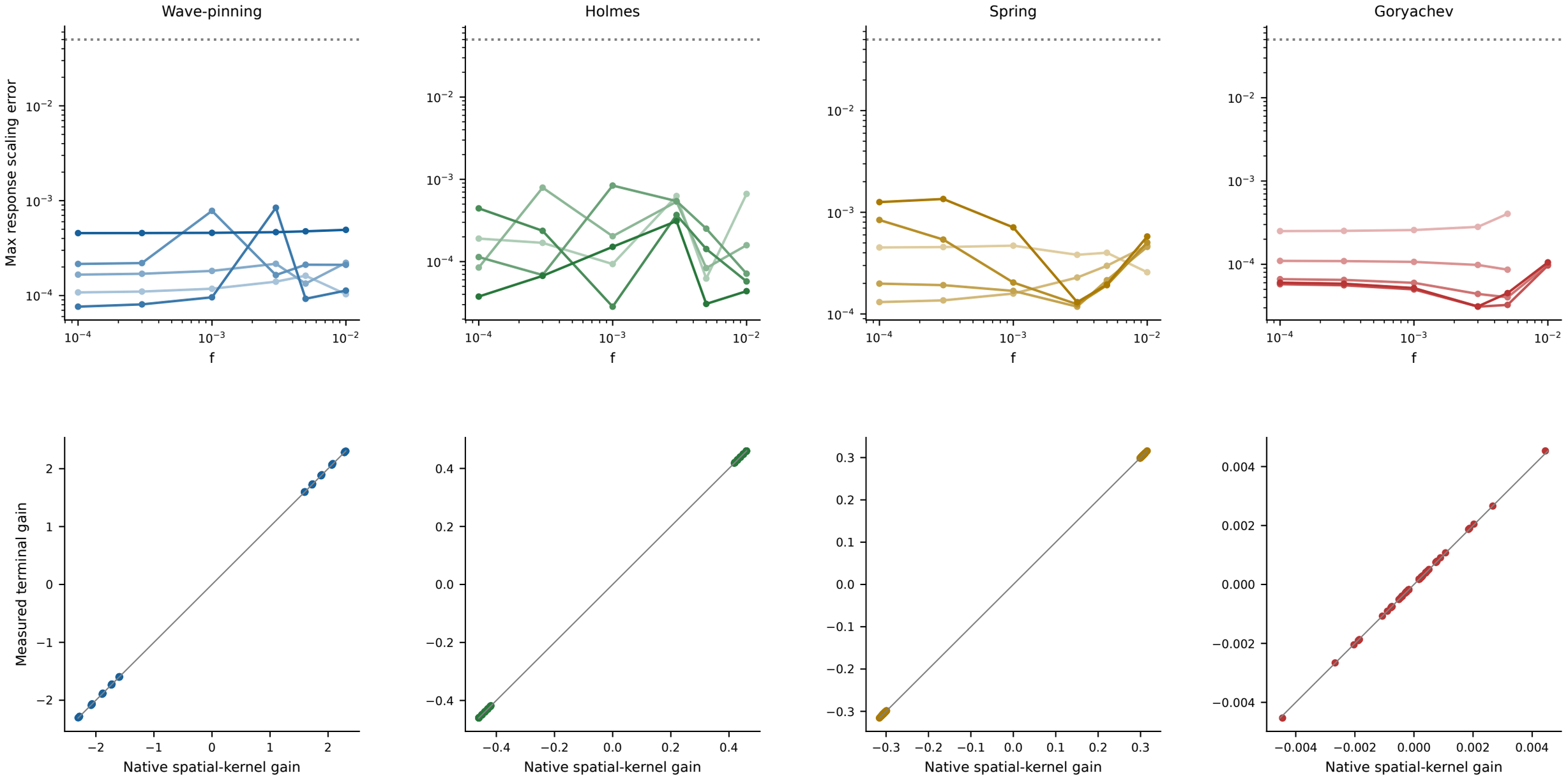
$$D_s^{\mathrm{ind}} = \tfrac{D_L + D_R}{4}, \qquad D_s = \tfrac{D_L + D_R + 2D_{LR}}{4}. \tag{5}$$

$$\mathcal{A}_{20\tau} = \tfrac{1}{a\tau}\int_0^{20\tau} [x_e^{\mathrm{pulse}}(t) - x_e^{\mathrm{control}}(t)]\,\mathrm{d}t. \tag{6}$$

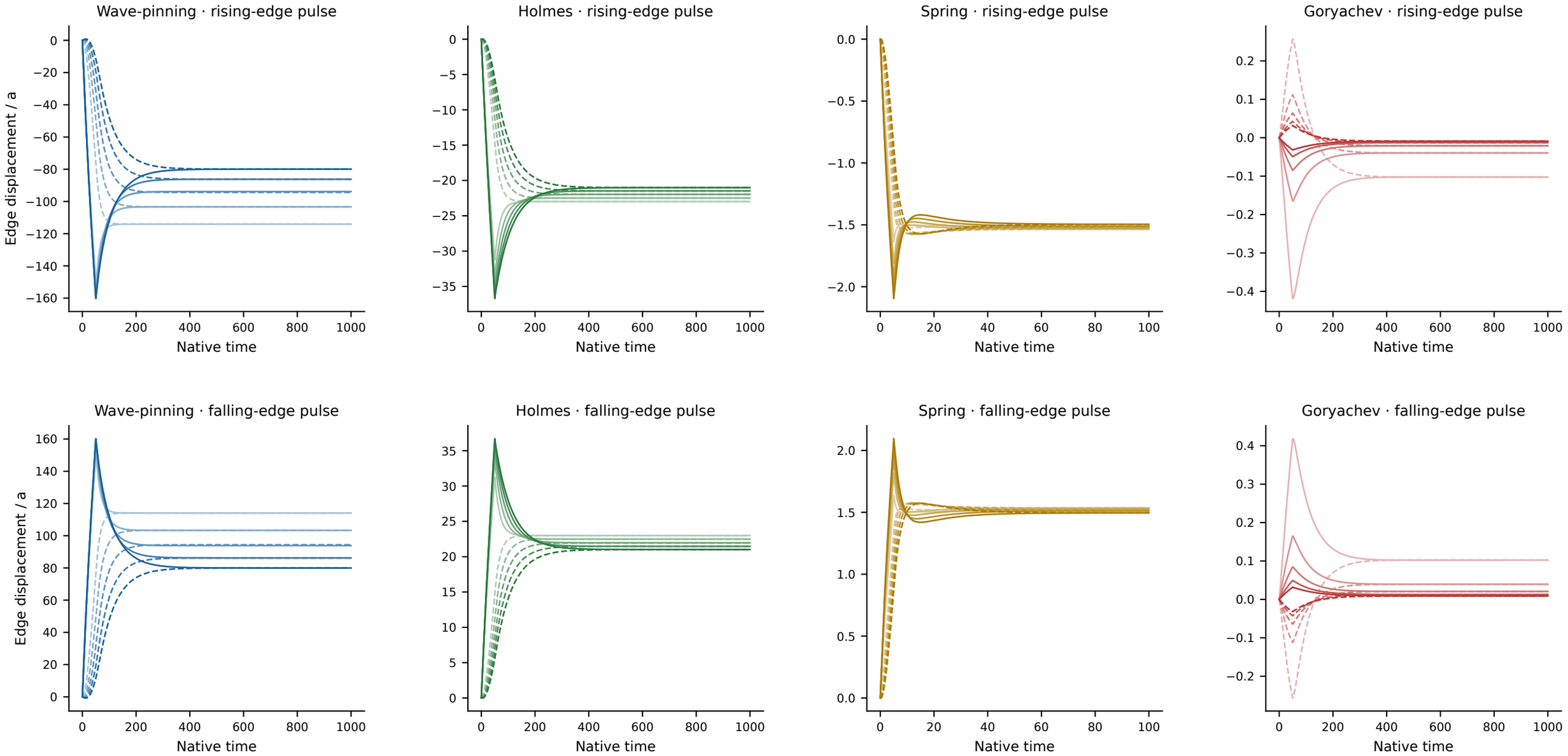
$$\widehat{g}_{e,j} = \tfrac{M_e}{Q_e} g_j, \quad M_e = \tfrac{\Delta x_e(20\tau)}{a\tau}, \quad Q_e = \sum_j g_j h_{e,j}. \tag{7}$$

q: derivative of the translated native profile; p: its adjoint, excluding conserved-pool modes.
Cx: discrete circular filter autocorrelation. Ct: analytically variance-normalized OU-knot interpolation.
The overbar averages the knot phase. Native sigma = 1; lognormal input covariance is not Gaussian OU covariance.

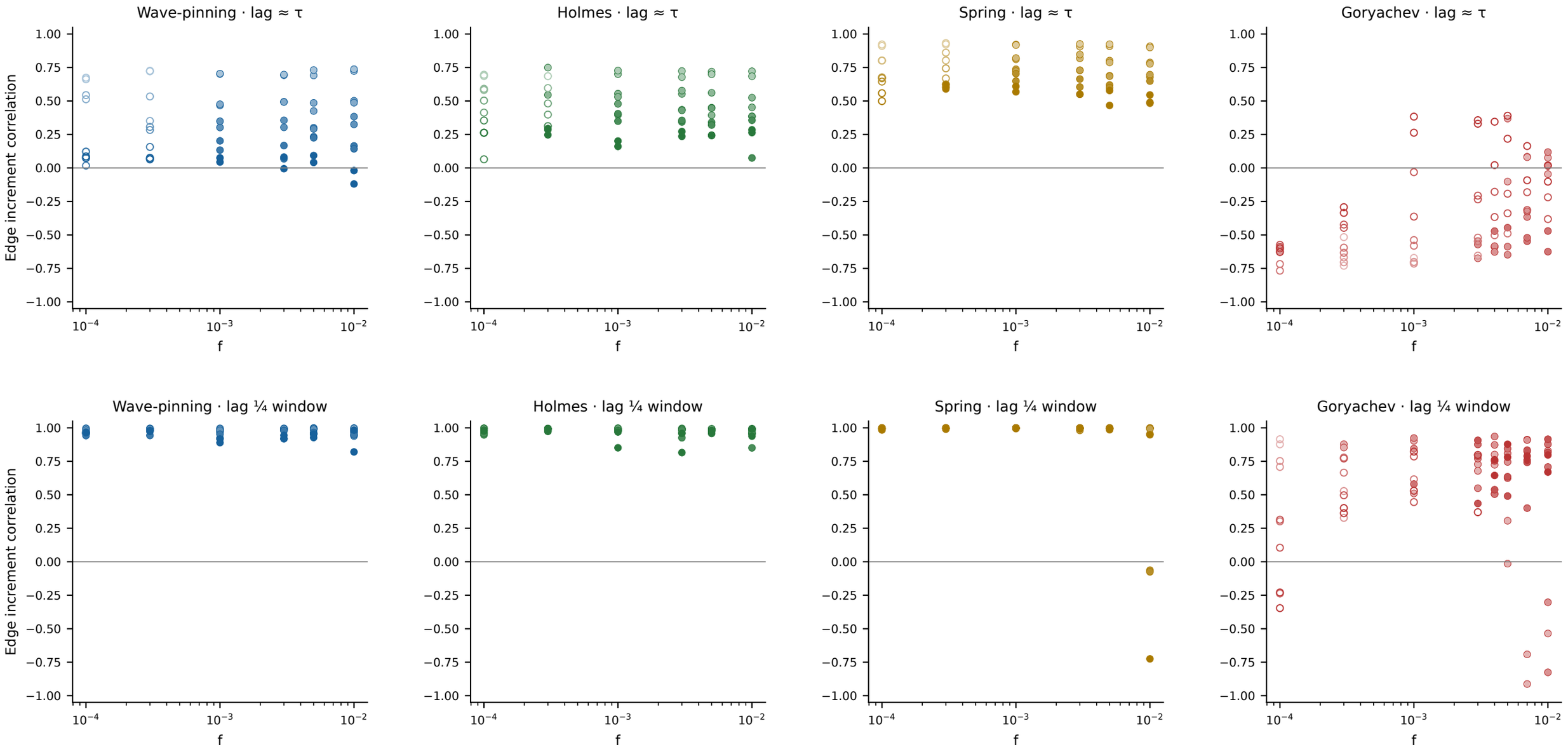
Amplitude checks and pulse validation; neither uses observed drift



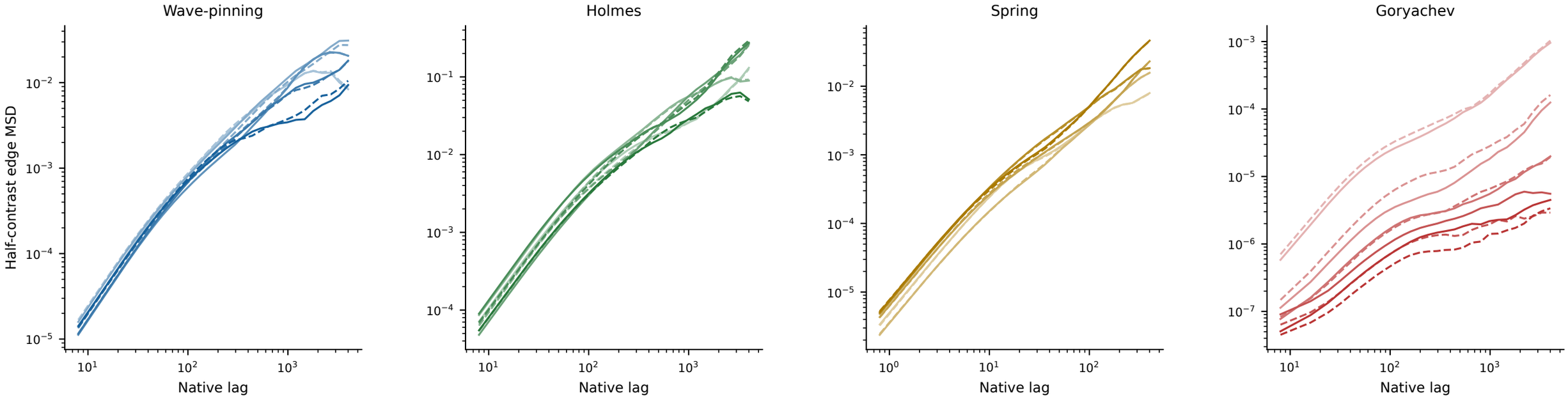
Both half-contrast edges move after a local pulse; f = 0.001



Solid: pulsed edge; dashed: opposite edge. Each shade is a Figure 4 size. A persistent common displacement is the neutral translation mode.

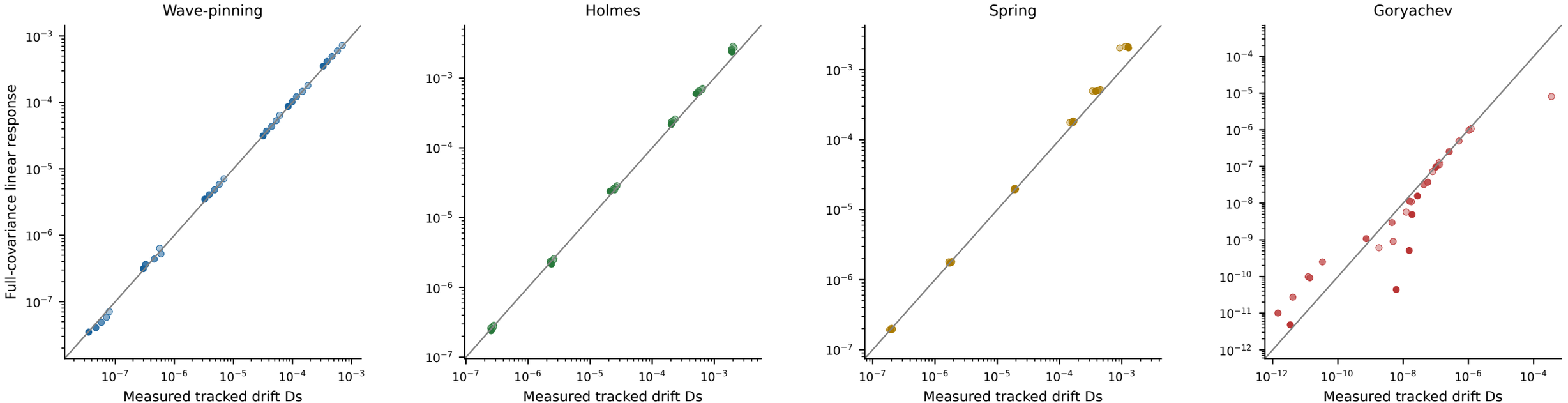

Independence check: Figure 4 exemplar seeds 0 and 1
Wave-pinning · lag ≈ τ
Holmes · lag ≈ τ
Spring · lag ≈ τ
Goryachev · lag ≈ τ
Wave-pinning · lag ¼ window
Holmes · lag ¼ window
Spring · lag ¼ window
Goryachev · lag ¼ window
Edge increment correlation
f
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
10^{-4}
10^{-3}
10^{-2}
Filled: both edge MSDs >10× conservative float16 crossing-error bound²; open: saved-profile precision may dominate.
Each point is a seed/size/input, only uninterrupted single-region pairs. These two exemplars do not give campaign confidence intervals.

Edge MSDs at f = 0.001; solid rising, dashed falling, seed 0

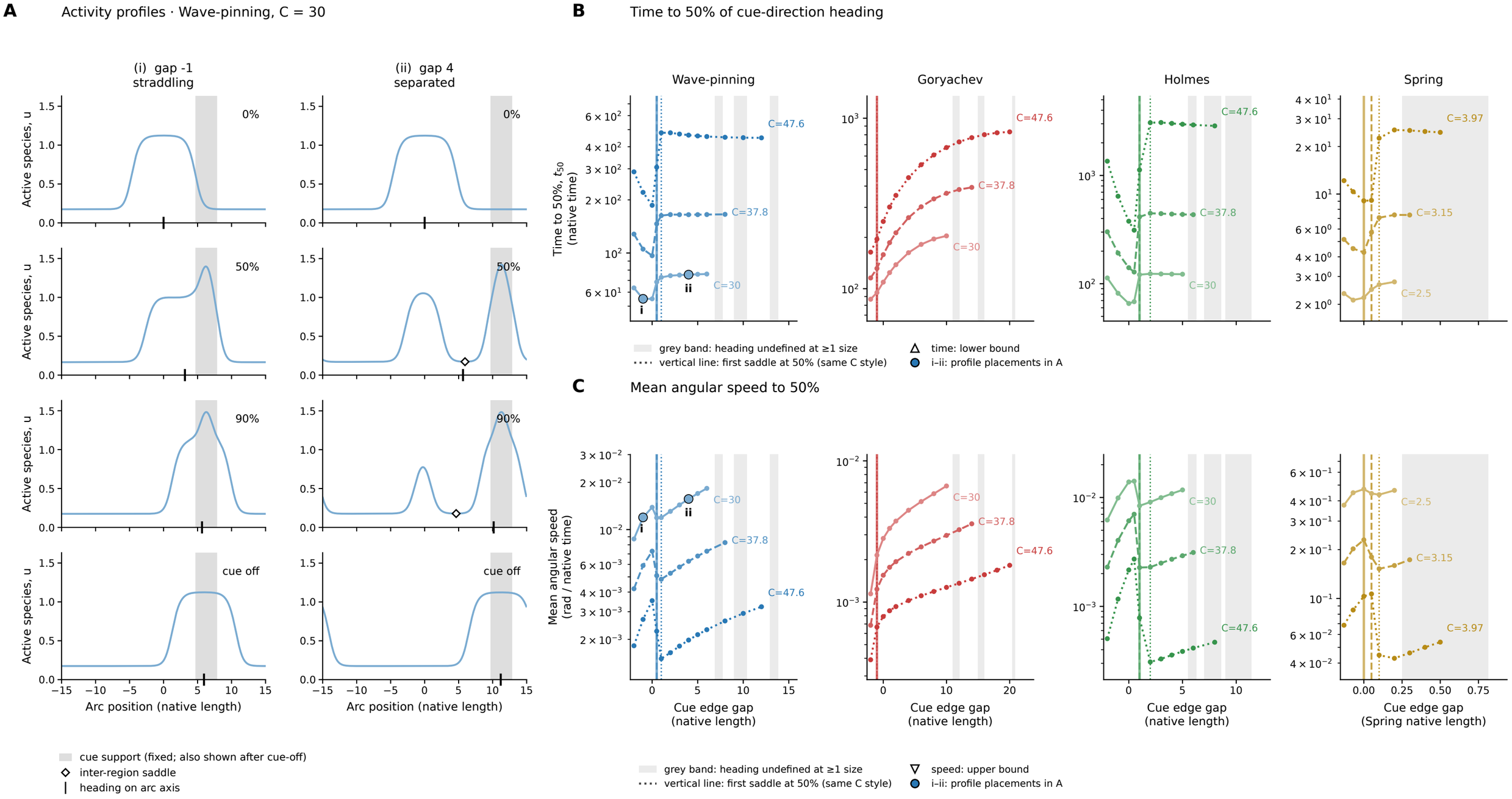


All seeds, fractions, pair counts, quantization bounds and covariance factors are provided in the plotted-data tables.

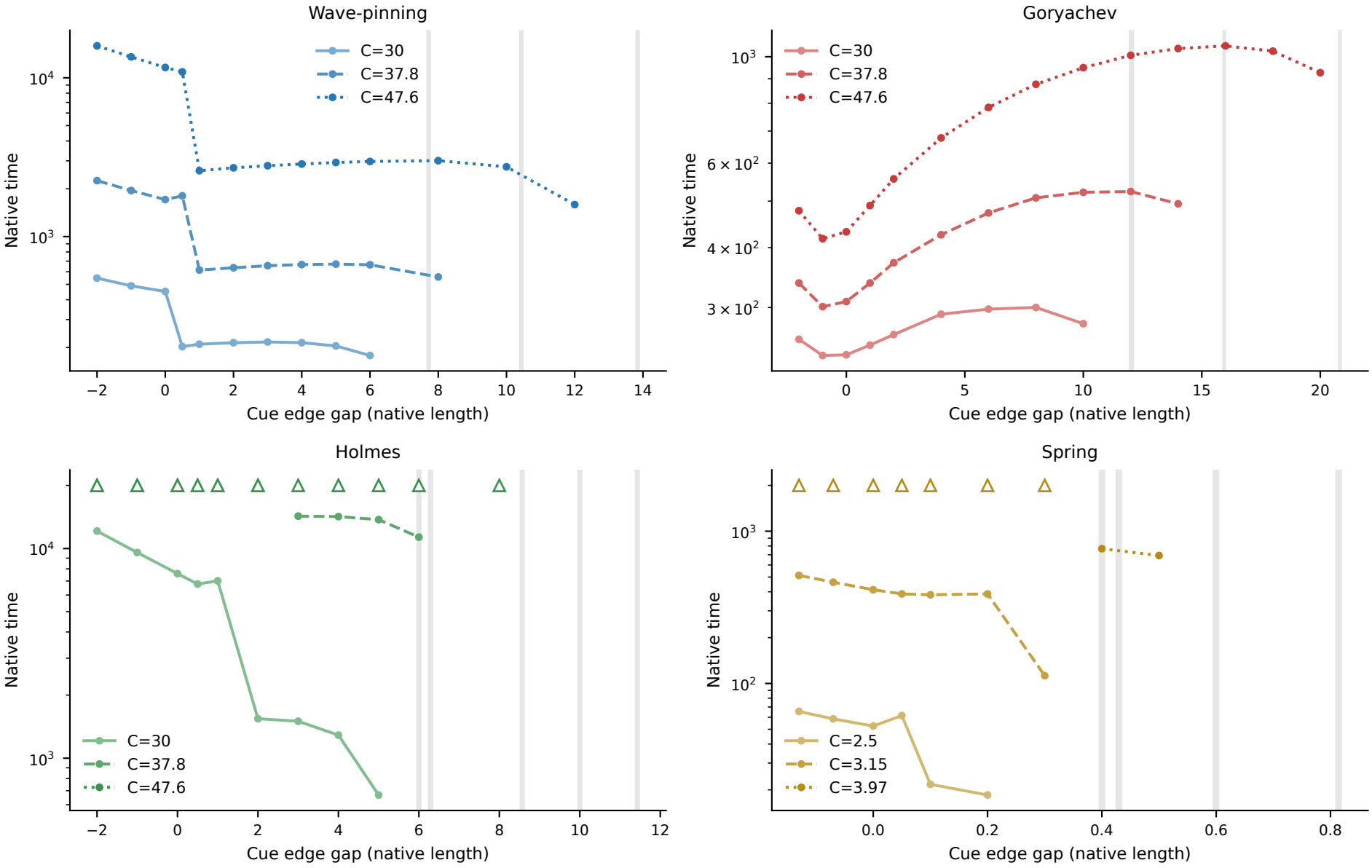
Full covariance sensitivity: retain the shared neutral translation



No parameter fit: the full correlated prediction is approximately twice the prescribed independent-edge prediction because the two measured-gain kernels nearly coincide.
This is a prediction of physical tracked drift, not an explanation of all heading decay, especially for Goryachev.

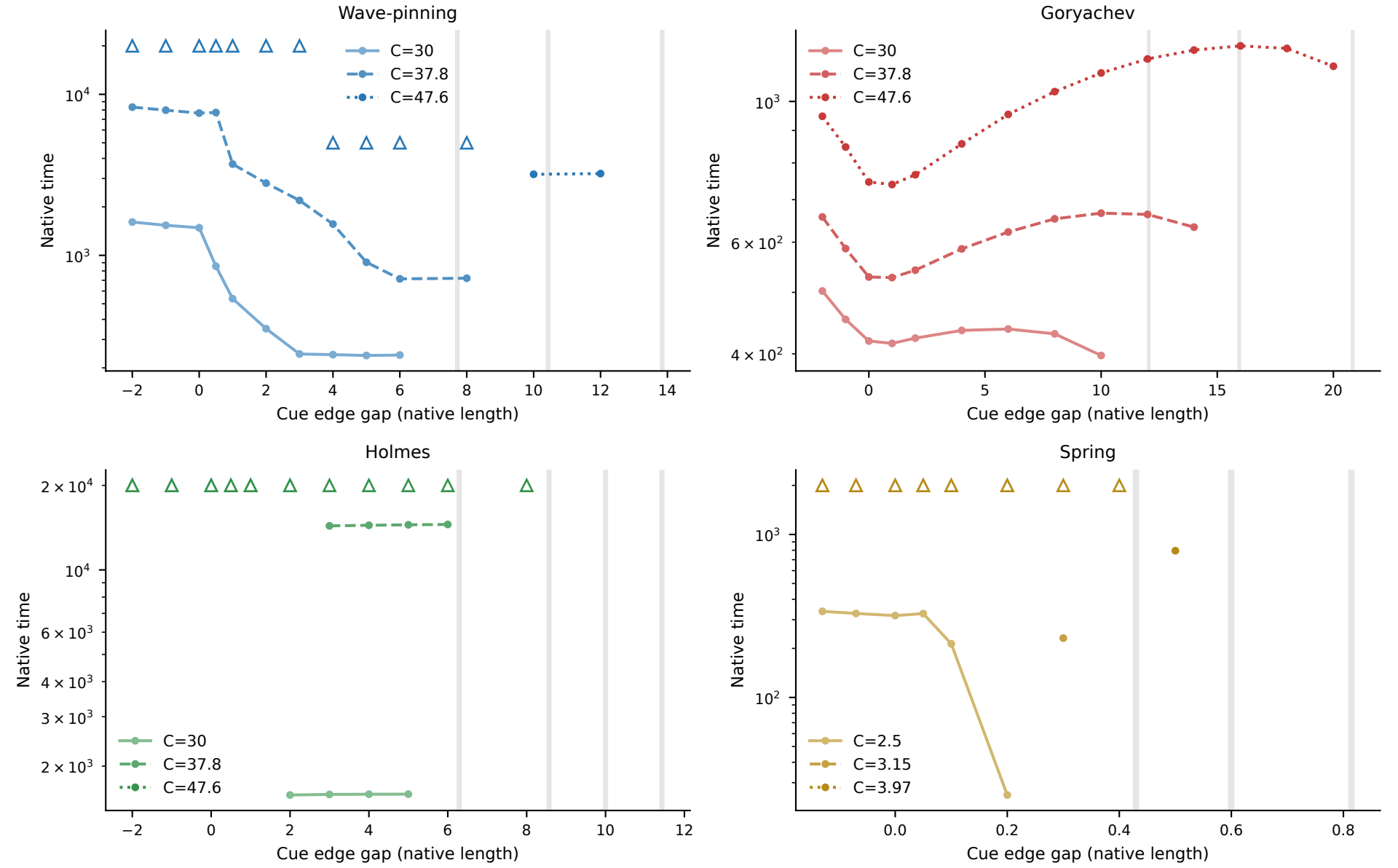

A
Activity profiles · Wave-pinning, C = 30
(i) gap -1
straddling
(ii) gap 4
separated
0%
50%
90%
cue off
Active species, u
Arc position (native length)
cue support (fixed; also shown after cue-off)
inter-region saddle
heading on arc axis
B
Time to 50% of cue-direction heading
Wave-pinning
Goryachev
Holmes
Spring
Time to 50%, t_{50}
(native time)
C=47.6
C=37.8
C=30
C=3.97
C=3.15
C=2.5
i
ii
grey band: heading undefined at ≥1 size
vertical line: first saddle at 50% (same C style)
time: lower bound
i–ii: profile placements in A
C
Mean angular speed to 50%
Mean angular speed
(rad / native time)
Cue edge gap
(native length)
Cue edge gap
(Spring native length)
speed: upper bound

Supplement: time to 90% of target heading; continuous cue



Open up-triangles: lower bounds; grey: undefined heading. Midpoints of saved brackets; quality gate R ≥ 0.05. No cap is a measured time.

Supplement: time to 99% of target heading; continuous cue



Open up-triangles: lower bounds; grey: undefined heading. Midpoints of saved brackets; quality gate R ≥ 0.05. No cap is a measured time.

A Activity profiles · Wave-pinning

cue support (fixed; also shown after cue-off) ◇ inter-region saddle | heading on arc axis

(i) C = 30, gap -1 straddling | (ii) C = 30, gap 4 separated | (iii) C = 47.6, gap -1 straddling | (iv) C = 47.6, gap 4 separated

Active species, u

0% | 50% | 90% | cue off

Arc position (native length)

B Time to 50% of cue-direction heading

Wave-pinning | Goryachev | Holmes | Spring

Time to 50%, t_{50} (native time)

C=47.6 | C=37.8 | C=30 | C=3.97 | C=3.15 | C=2.5

i | ii | iii | iv

grey band: heading undefined at ≥1 size

vertical line: first saddle at 50% (same C style)

△ time: lower bound

i–iv: profile placements in A

C Mean angular speed to 50%

Mean angular speed (rad / native time)

Cue edge gap (native length)

Cue edge gap (Spring native length)

grey band: heading undefined at ≥1 size

vertical line: first saddle at 50% (same C style)

△ time: lower bound

▽ speed: upper bound

i–iv: profile placements in A

Figure 3. Cue placement relative to the incumbent’s edge switches reorientation between sliding the incumbent and takeover by a new region; enlargement slows both.

A. Native Wave-pinning active species u versus arc position centred on the incumbent’s initial centre, for (i) C = 30, gap −1; (ii) C = 30, gap 4; (iii) C = 47.6, gap −1; and (iv) C = 47.6, gap 4. Rows show 0%, 50% and 90% of the target heading under continuous cue, then the final cue-free profile. Grey shading marks the fixed cue support in every row; open diamonds mark an inter-region saddle where resolved, and black ticks on the arc axis mark heading. Each column shares its y-scale.

B. Time to 50% of the angle from initial activity-vector heading to the cue direction versus initial cue edge gap (near edge of cue support minus the incumbent’s 10%-contrast boundary). C. Mean angular speed over that turn, equal to half the target angle divided by t_{50}. Time and length use each model’s native units; speed is radians per native time. Spring retains its native mechanical units. Lighter solid, intermediate dashed and darker dotted curves denote increasing C, labelled at each line end. Enlarged filled i–iv markers identify A in both Wave-pinning plots. Both readouts are evaluated with the cue continuously on; points interpolate the saved 50% crossing brackets.

Vertical lines mark the first tested gap with two resolved regions and a saddle at the 50% crossing for each C, using that size’s line style; the existing periodic watershed detector is applied to the native saved-field interpolant. Light grey bands mark sampled near-antipode placements with undefined heading at one or more sizes (activity-vector coherence $R < 0.05$ or a cancellation interval); that size’s curve breaks there. Open up-triangles mark censored time lower bounds, and matching open down-triangles mark speed upper bounds; horizon caps are not measured values.

For the relaxation row only, cue-off occurs at 95% heading or at 20,000 if 95% is not reached, restarting from the full native event state after an identical cue-on replay. No-cue relaxation stops when $\max |\partial u/\partial t|$ reaches the existing tolerance 2.2683×10^{-7} (native activity per native time; the original 10^{-4} scaled-RHS tolerance over 1,000 time units), or after four times the cue-on duration. i: cue off at 95% heading, t = 790.5; relaxed for 258.6, stopped at stationarity; ii: cue off at 95% heading, t = 227.6; relaxed for 291.3, stopped at stationarity; iii: cue off at the 20,000 horizon (95% not reached), t = 20000.0; relaxed for 437.4, stopped at stationarity; iv: cue off at 95% heading, t = 2880.0; relaxed for 496.2, stopped at stationarity.

deBelly is excluded from rings.

Cellular measurement motivated by this assay: photoactivation at varying offsets from the leading-edge boundary in WGD+ versus WGD− cells, with time to half-turn as the readout.

Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution.

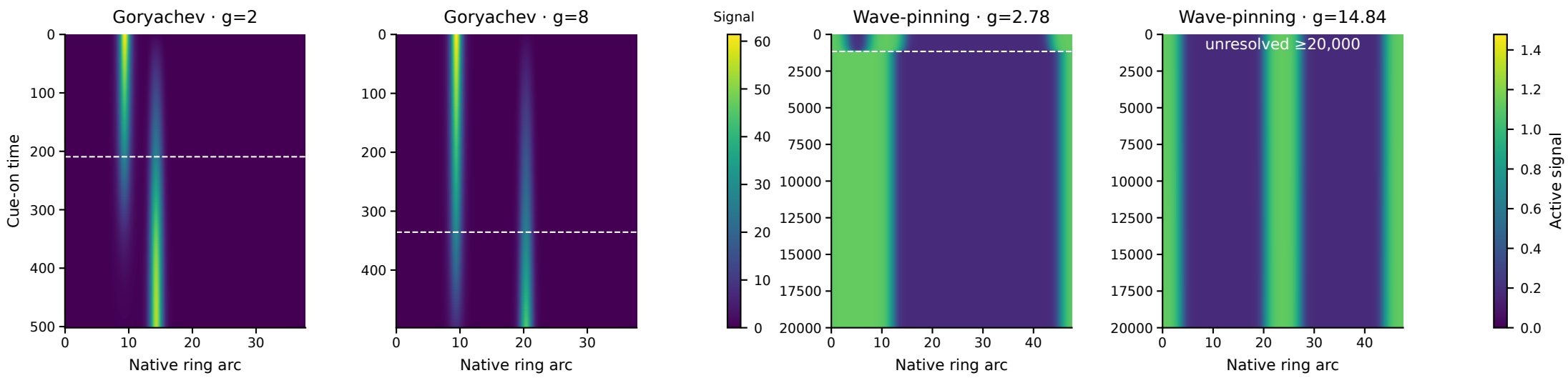
A Rotating a cue separates distance from native cell size



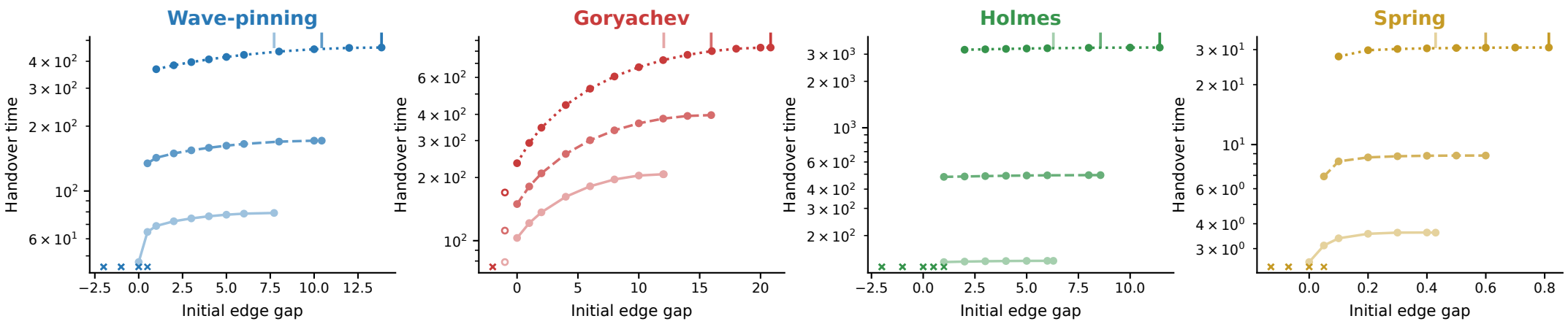
Same physical gap across sizes

A1: continuous cue → handover
B: equal complete-region amounts,
cue withdrawal → single-peak resolution
Blue incumbent; orange cue; g is edge gap.

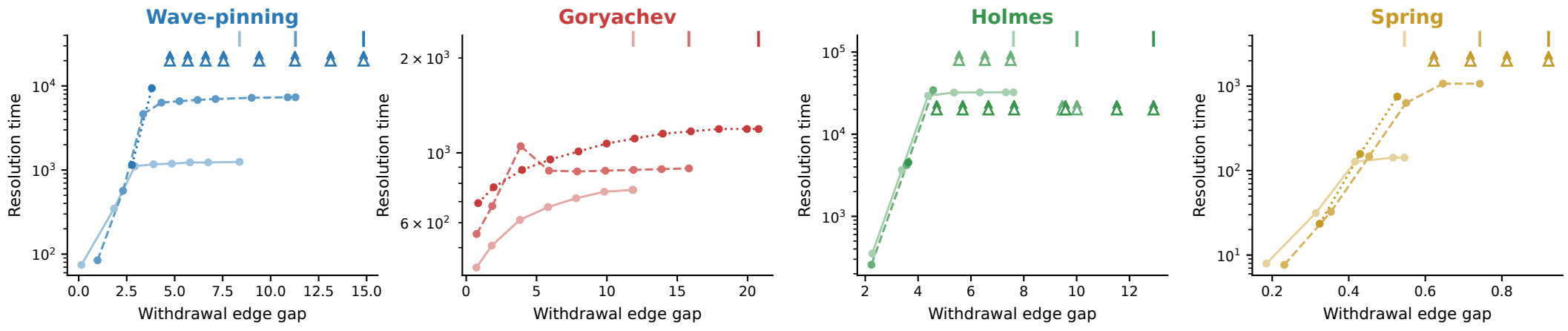
B Native histories: driven handover (left); time after withdrawal (right)



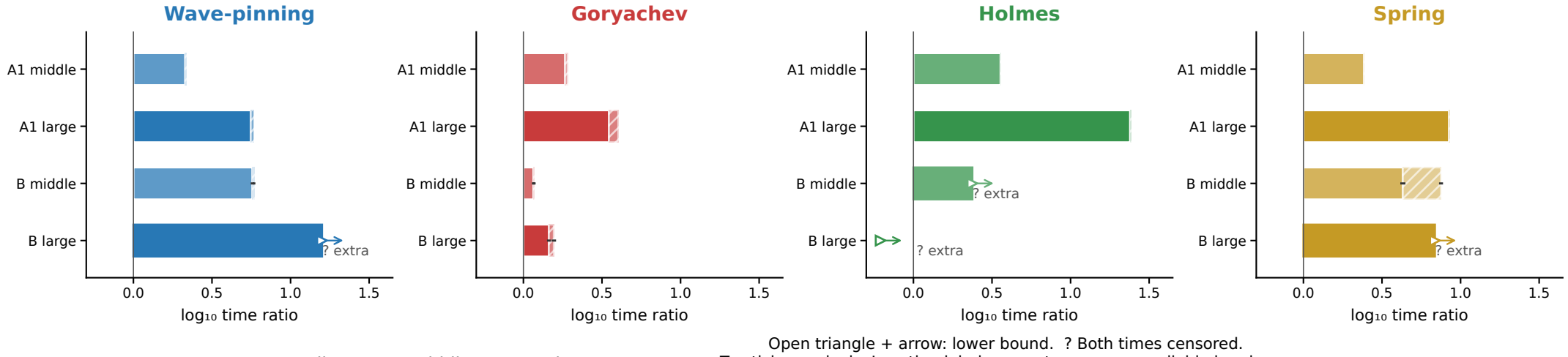
C Driven handover against initial edge gap



D Cue-free resolution against measured withdrawal gap · equal amounts



E Size at matched gap (solid) + extra separation (hatched)



Open triangle + arrow: lower bound. ? Both times censored.
Top ticks: each size's antipodal-placement gap. ×: unavailable handover.

Figure 3 round 2 · decomposition and censoring

Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution. Extra separation adds at most 1.15-fold to driven handover. In resolved cue-free comparisons it adds about 1.01-1.07-fold in Wave-pinning and Goryachev; Spring middle size is the supported exception, about 1.75-fold. Censored comparisons do not establish a numerical extra-separation effect.

For each model and assay, the matched gap is the largest gap within both measured size curves. In this dataset it is the small-size antipodal-placement endpoint. In B, the coordinate is the measured gap at withdrawal, not the initial cue gap. The endpoint called antipode is the trajectory seeded by the initial antipodal cue: its withdrawal region centers can shift.

A1 uses native continuous complete-region handover roots. B uses saved-sample lower and upper time brackets. Linear interpolation is restricted to adjacent measured eligible placements. Interval endpoints are interpolated separately; any censored upper endpoint stays infinite. Missing or topology-ineligible values are never converted into bounds.

A ratio with a censored numerator and resolved denominator has a lower bound. A resolved numerator and censored denominator gives an upper bound. Two censored times give no finite one-sided ratio bound; the result is marked indeterminate. In particular, dividing two horizon caps is invalid. Finite bars use the geometric center of the resulting interval, with the full interval recorded in CSV.

The exact product identity uses the same interpolated matched-gap time in both terms. For finite B intervals the representative values obey the identity; multiplying independently propagated marginal interval limits would ignore this shared denominator and numerator. Total intervals are computed directly from endpoint times.

All 9 author-approved Holmes ratio-1 extensions (5 small, 4 middle) completed to 80,000 from exact archived 20,000 endpoints. Six resolve and three remain censored. Other Holmes conditions retain their original 20,000 horizon; different lower bounds reflect observation duration, not a decline in resolution time. No cue, kinetic parameter, mesh or resolution definition changed.

The following pages retain BOTH superseded main pages, including the complete translation page, then the entire seven-page existing supplement. All four native-profile appendix volumes remain linked in the existing reader; no model inputs, histories, or prior supplements are replaced.

Approved Holmes cue-free extensions: every trajectory continued to 80,000

Circumference	Withdrawal gap	Resolution time	Final native state	Field steady
30	4.3786	29,420–29,440	single region incumbent associated	True
30	5.3610	32,160–32,180	single region incumbent associated	True
30	6.3425	32,280–32,300	single region incumbent associated	True
30	7.3241	32,320–32,340	single region incumbent associated	True
30	7.6088	32,320–32,340	single region incumbent associated	True
37.8	4.5702	34,380–34,400	single region incumbent associated	True
37.8	5.5498	≥80,000	two resolved spatially associated regions	False
37.8	6.5270	≥80,000	two resolved spatially associated regions	False
37.8	7.5037	≥80,000	two resolved spatially associated regions	True

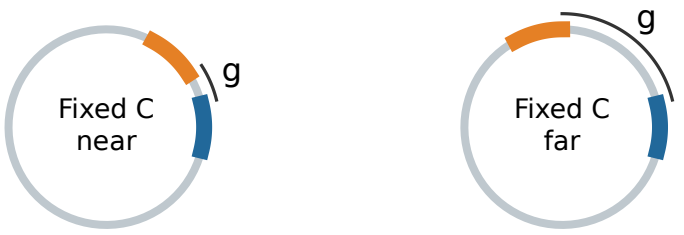
Full native fields and complete-region readouts are retained. Fate and full-field stationarity are independent.
Six observed resolutions; three one-sided lower bounds. Original campaigns are unchanged.
Mixed 20,000/80,000 bounds on the main curves identify observation duration, not point estimates.

All decomposition values · time ratios with interval bounds

Model	Assay	Size / small	Matched gap	Size term	Extra separation	Total
Wave-pinning	A1	37.8 / 30	7.719	2.14	1.01	2.17
Wave-pinning	A1	47.6 / 30	7.719	5.59	1.05	5.86
Goryachev	A1	37.8 / 30	12.05	1.84	1.04	1.91
Goryachev	A1	47.6 / 30	12.05	3.5	1.15	4.02
Holmes	A1	37.8 / 30	6.29	3.59	1	3.6
Holmes	A1	47.6 / 30	6.29	23.9	1.01	24.2
Spring	A1	3.15 / 2.5	0.4296	2.44	1	2.44
Spring	A1	3.97 / 2.5	0.4296	8.45	1.01	8.52
Wave-pinning	B	37.8 / 30	8.378	5.67–5.77	1.02–1.03	5.81–5.92
Wave-pinning	B	47.6 / 30	8.378	≥15.9	indeterminate (both censored)	≥15.9
Goryachev	B	37.8 / 30	11.87	1.15–1.16	1.01–1.02	1.16–1.18
Goryachev	B	47.6 / 30	11.87	1.44–1.47	1.06–1.09	1.54–1.58
Holmes	B	37.8 / 30	7.609	≥2.37	indeterminate (both censored)	≥0.618
Holmes	B	47.6 / 30	7.609	≥0.618	indeterminate (both censored)	≥0.618
Spring	B	3.15 / 2.5	0.5456	4.2–4.38	1.73–1.77	7.31–7.71
Spring	B	3.97 / 2.5	0.5456	≥6.87	indeterminate (both censored)	≥13.8

Intervals reflect saved-time brackets, not biological uncertainty. ≥ is a one-sided lower bound.
All interval endpoints, interpolation weights and source slots appear in the companion CSVs.

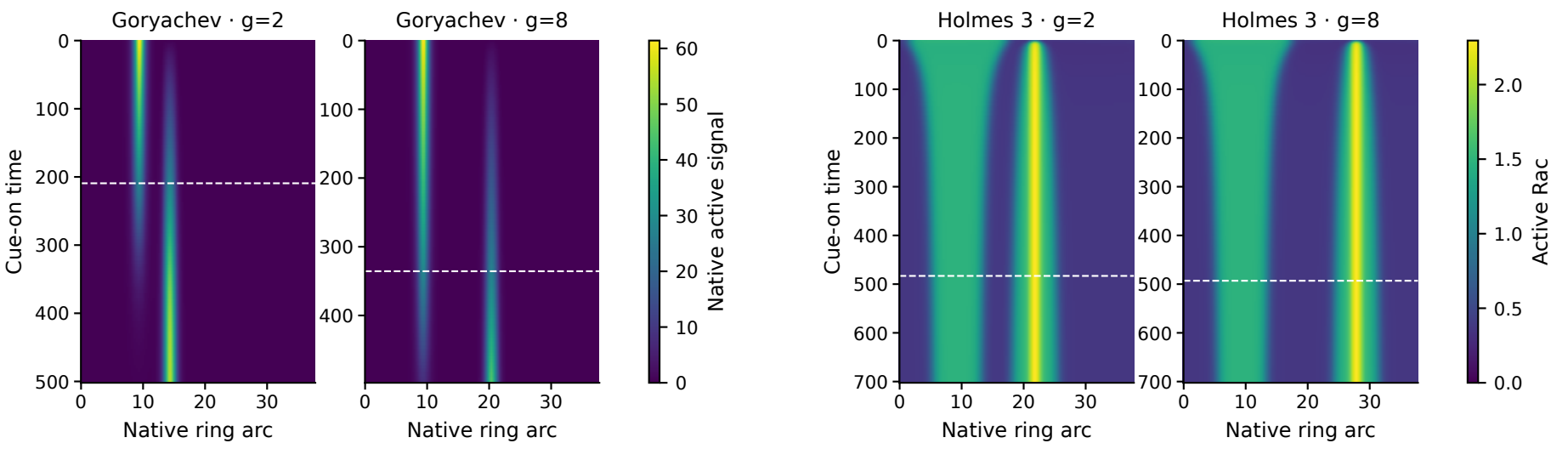
Figure 3 · Separation affects native competition differently across models



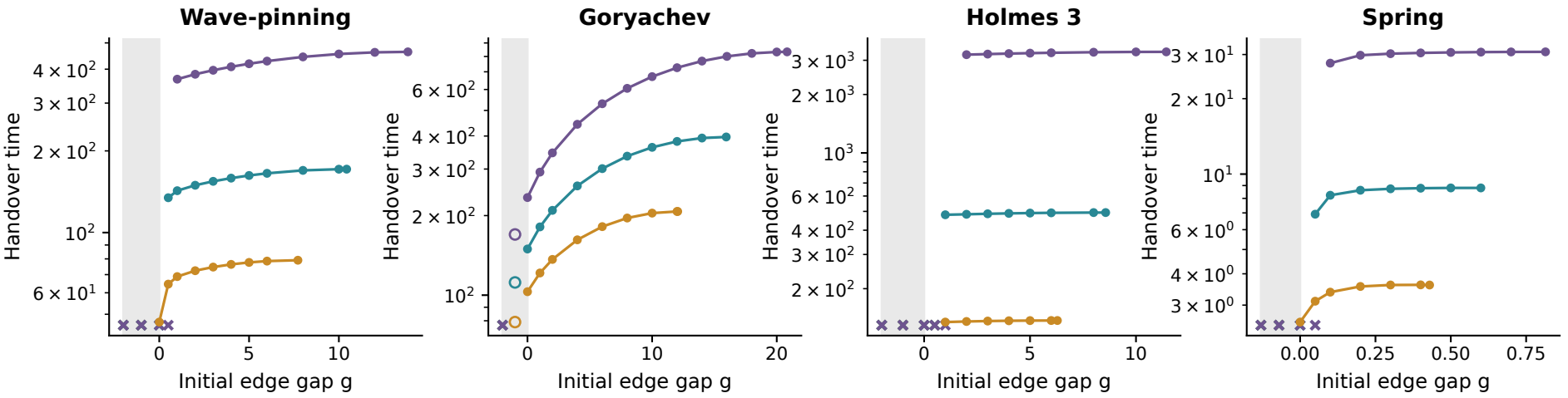
A Rotate one native cue at fixed size

g: initial incumbent 10% edge → cue near edge
A1: continuous cue → complete-region handover
A2: own-handover-scaled pulse → cue-free fate
B: seed ratio 0.5 / 1 / 2 → cue-free peak resolution
12 fresh qualified basals; native kinetics unchanged

B Same-size native histories · white line: complete-region handover



C1 Driven handover · open symbols: borderline placements



C2 Cue-free resolution · ratio 1 · triangles retain finite-cap lower bounds

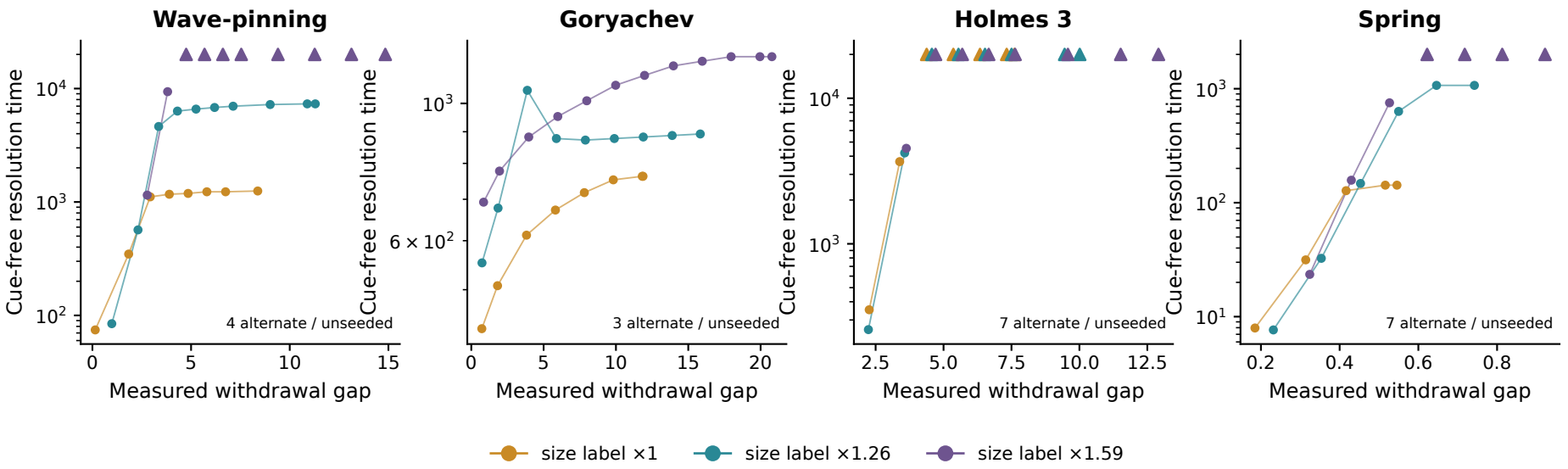


Figure 3 · Geometry remaps challenges; activity direction is measured separately

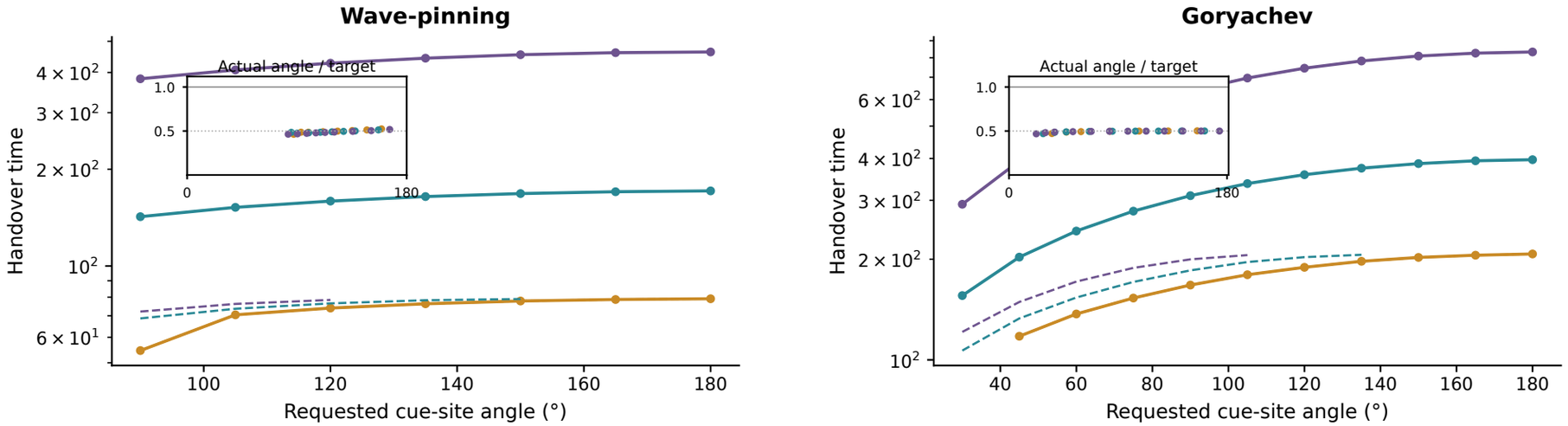


D Physical gap and center angle

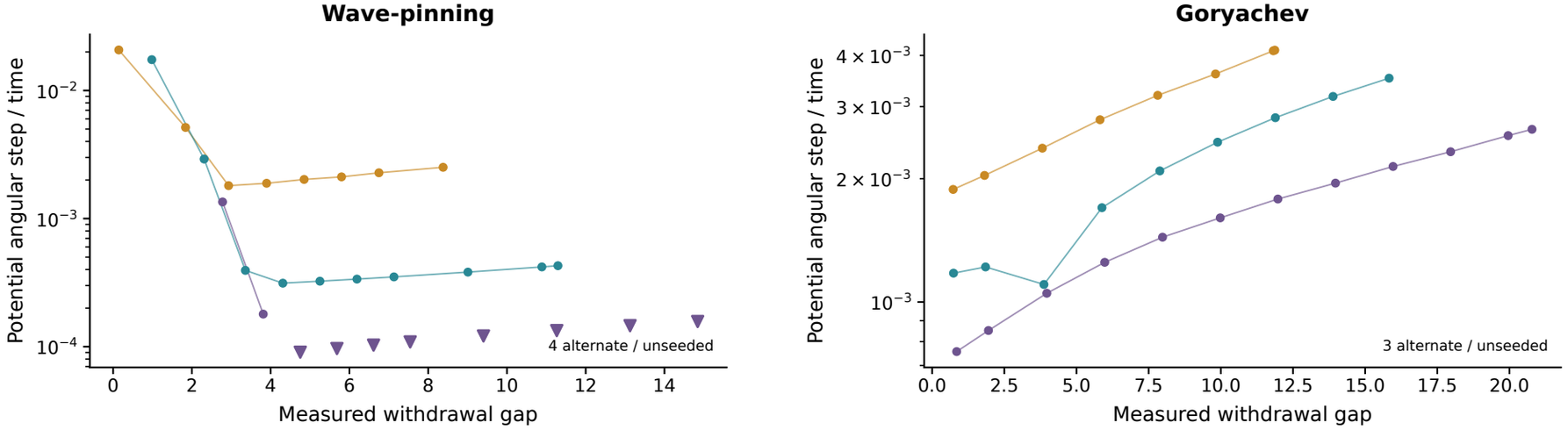
$$\theta = \frac{2\pi}{C}\left[g + \frac{w_{10} + s}{2}\right]$$

Measured widths enter the center step.
Handover and full-angle rotation are separate events.

E Fixed-angle cue placement: measured handover (solid), small-cell remap (dashed)

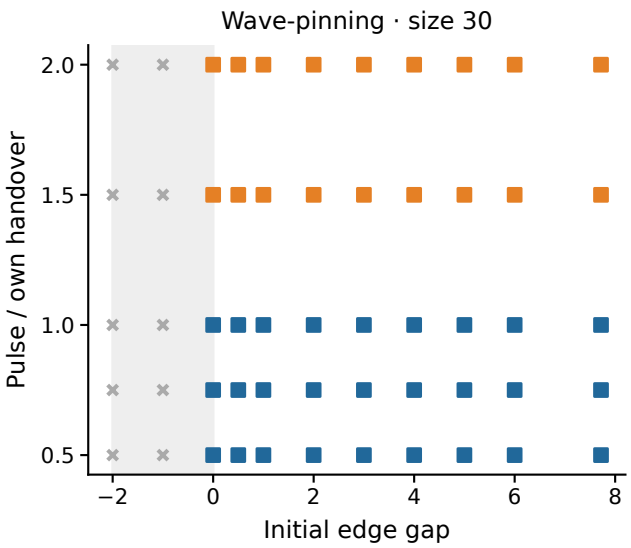


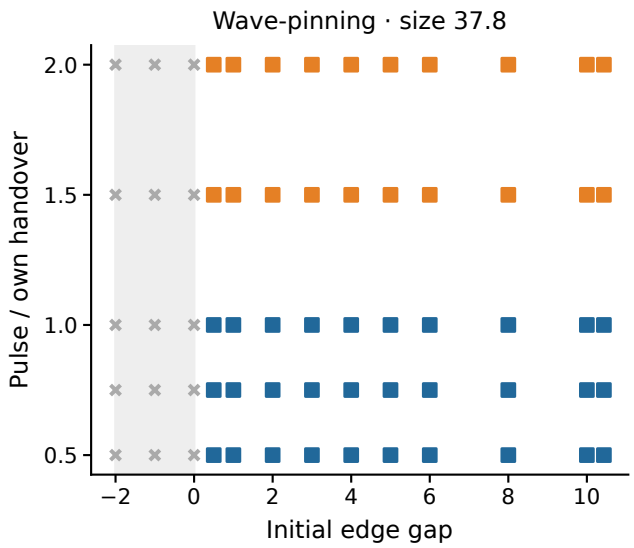
F Fixed physical scale: conditional geometric step / resolution time

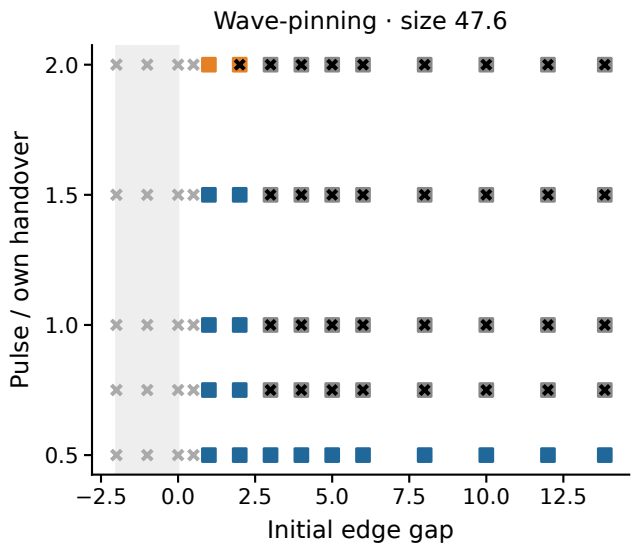


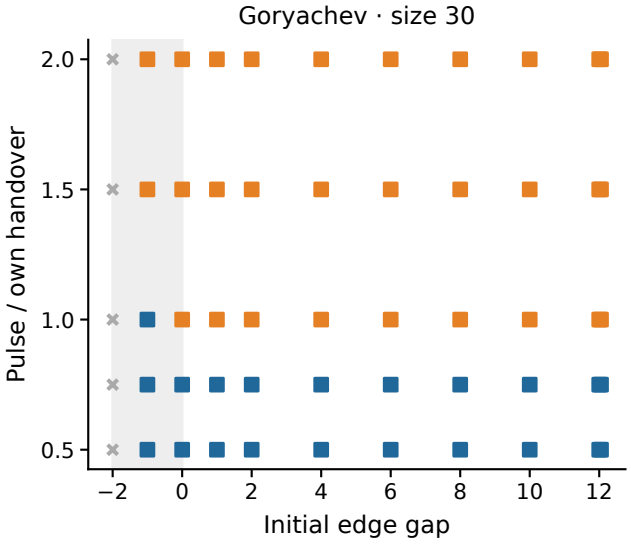
Dashed remaps stop at observed small-cell support. Insets measure actual global angle at handover, R≥0.05.
Potential step uses measured withdrawal centers and widths; direct heading-time tests and spatial outcomes are retained in the supplement.
These are native polarity-activity readouts; observed migration and conditional OU persistence estimates remain separate.

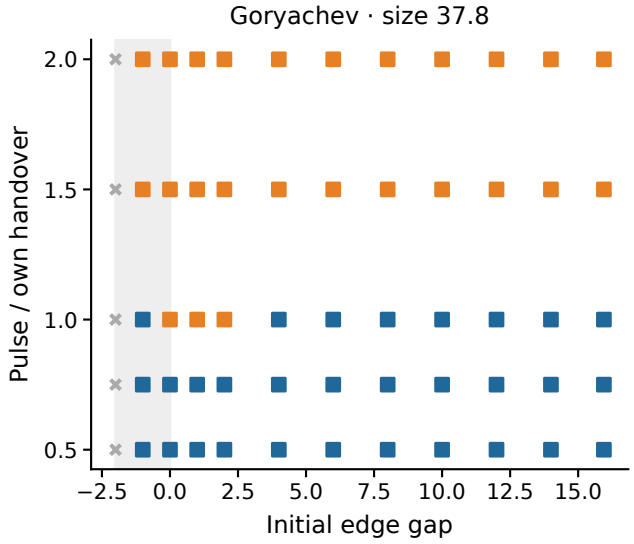
S1 · All 545 cue-withdrawal pulses and 34 unavailable own-handover placements

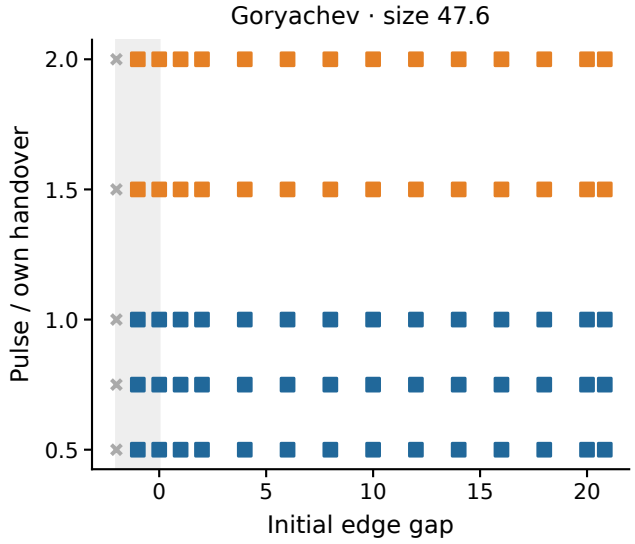


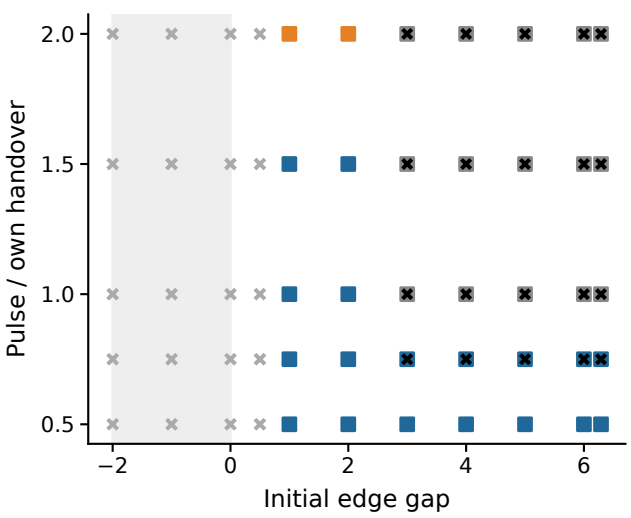


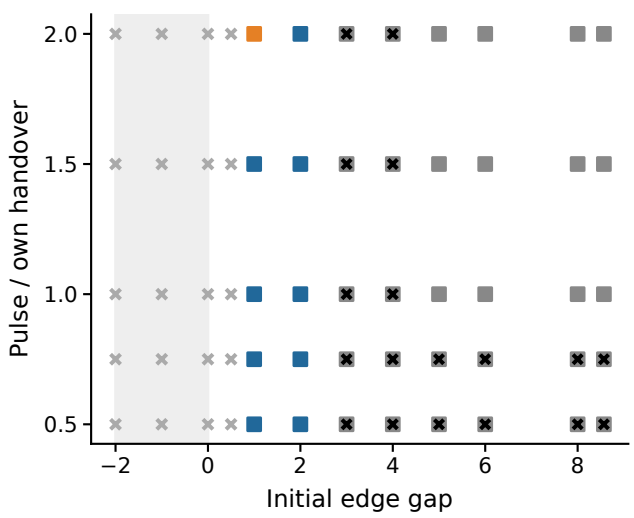


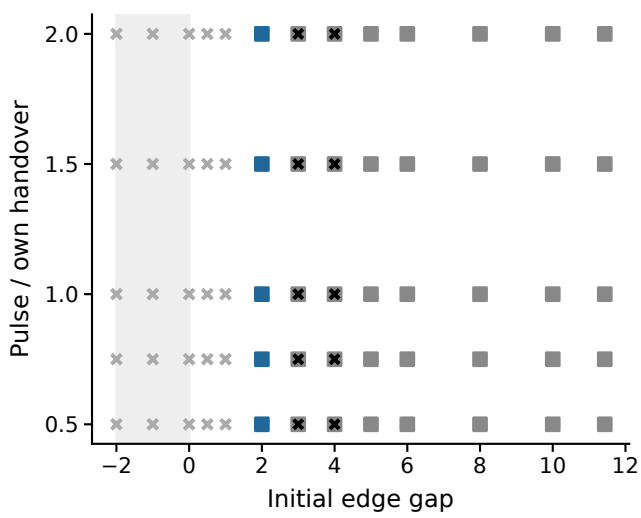


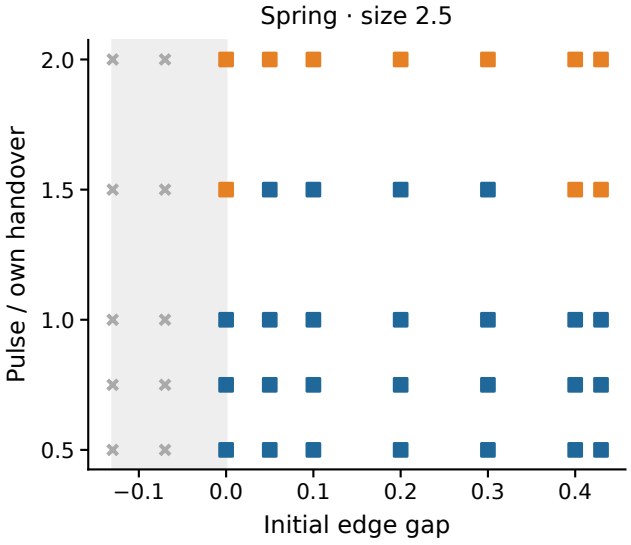


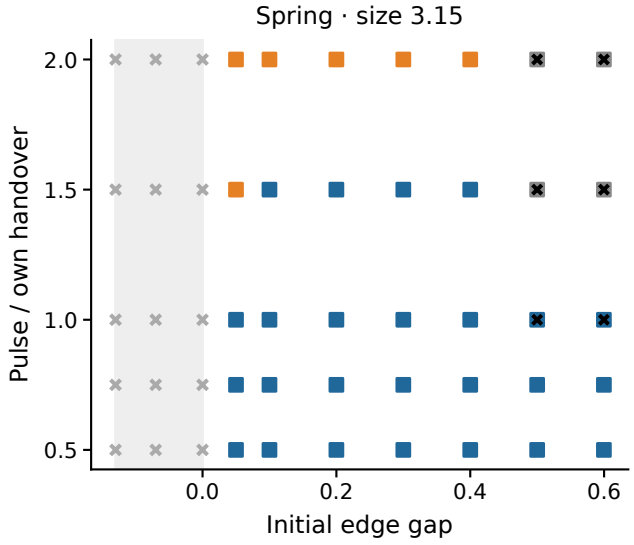


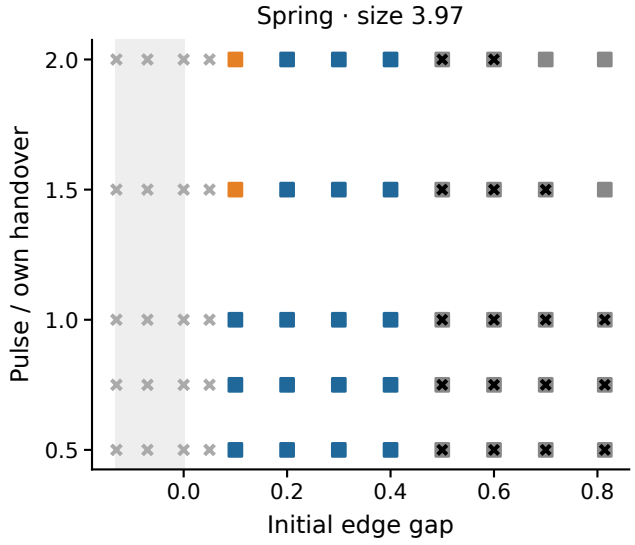








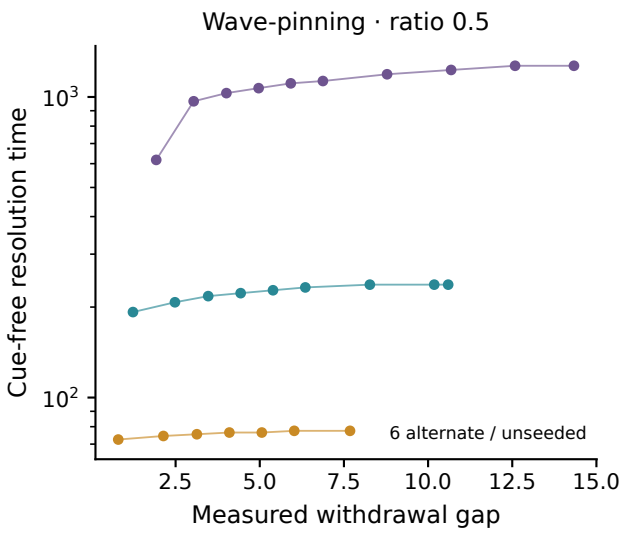


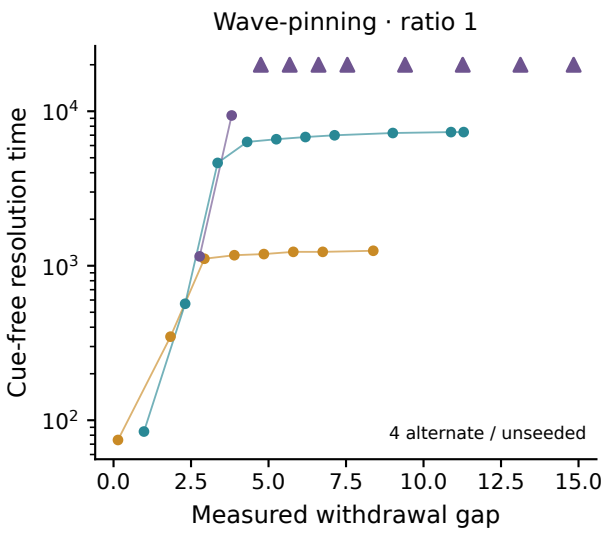


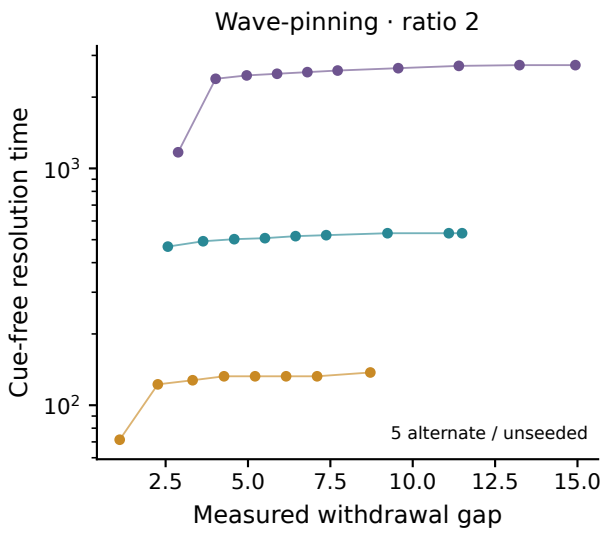
Blue: incumbent-associated final region; orange: challenger-associated; grey: unresolved. Grey ×: no true two-region handover.
Black × over a tested cell: finite native-field tail fails, independent of spatial fate. Shading: borderline placement.

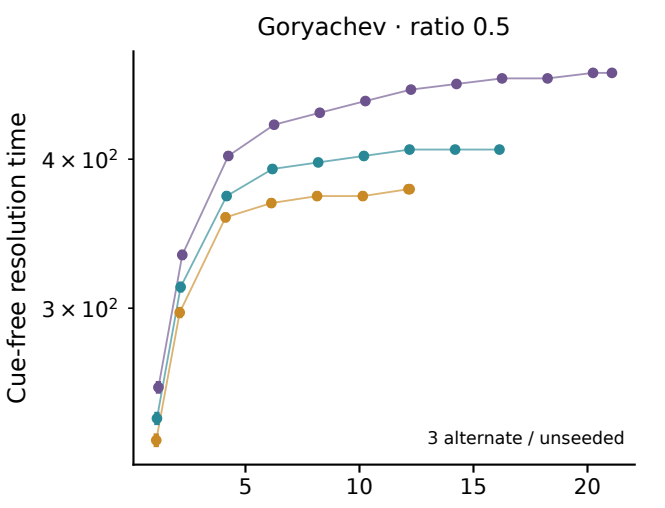
S2 · Cue-free resolution at all three complete-region withdrawal ratios

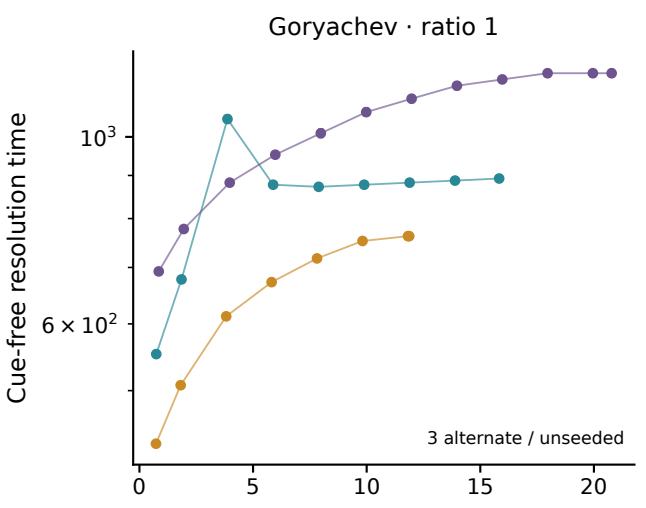
reference size · 1.26× native size label · 1.59× native size label

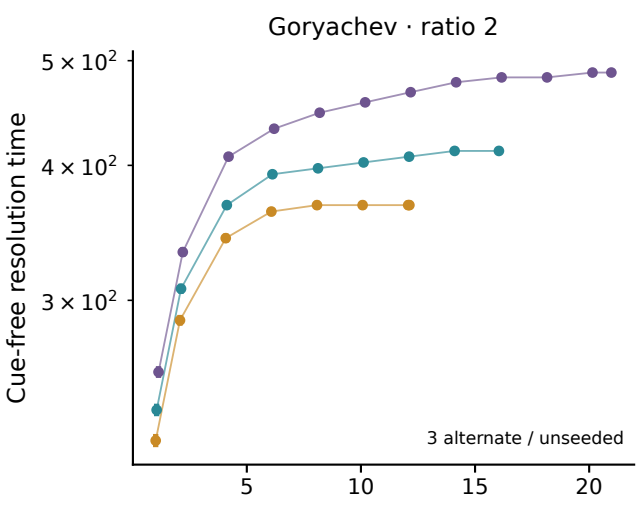


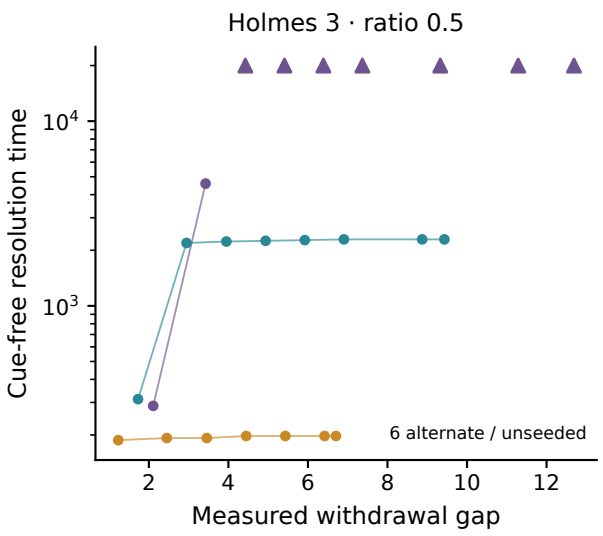


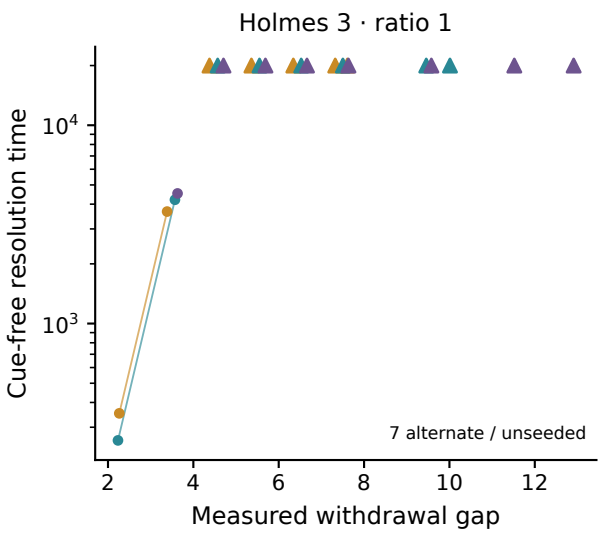


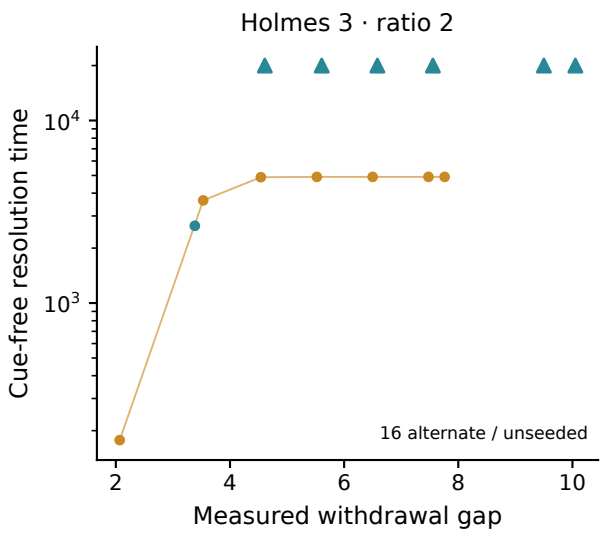


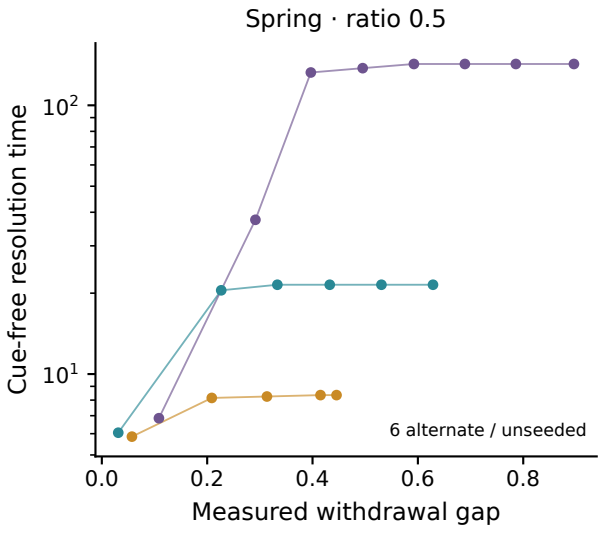


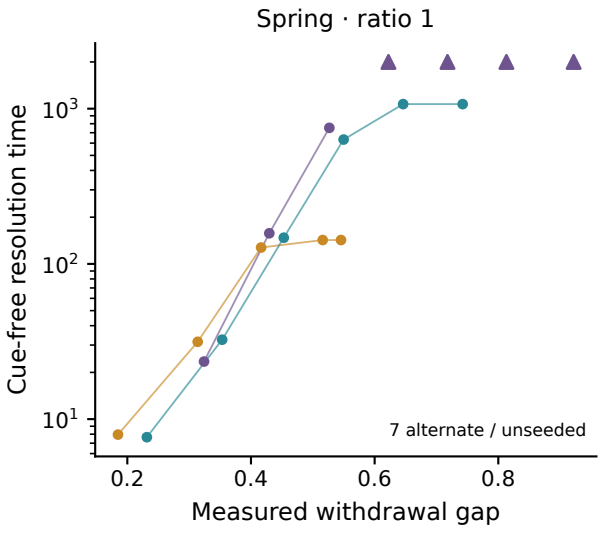


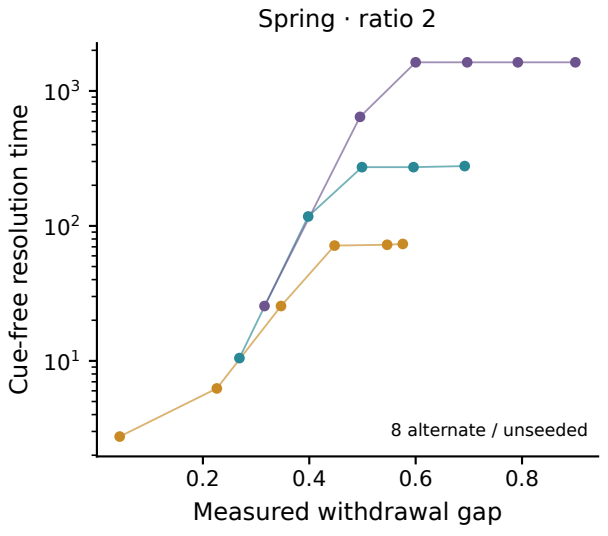




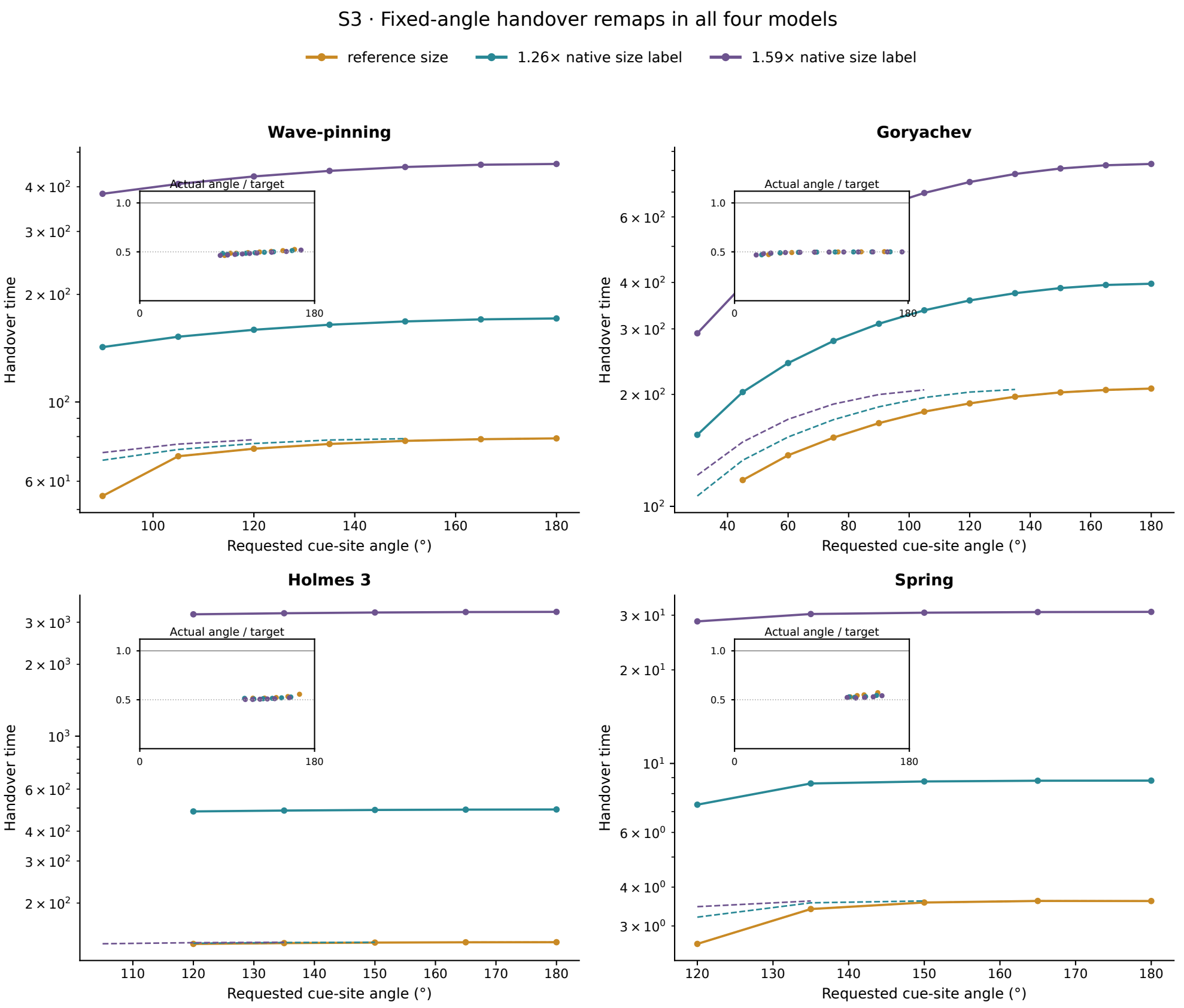








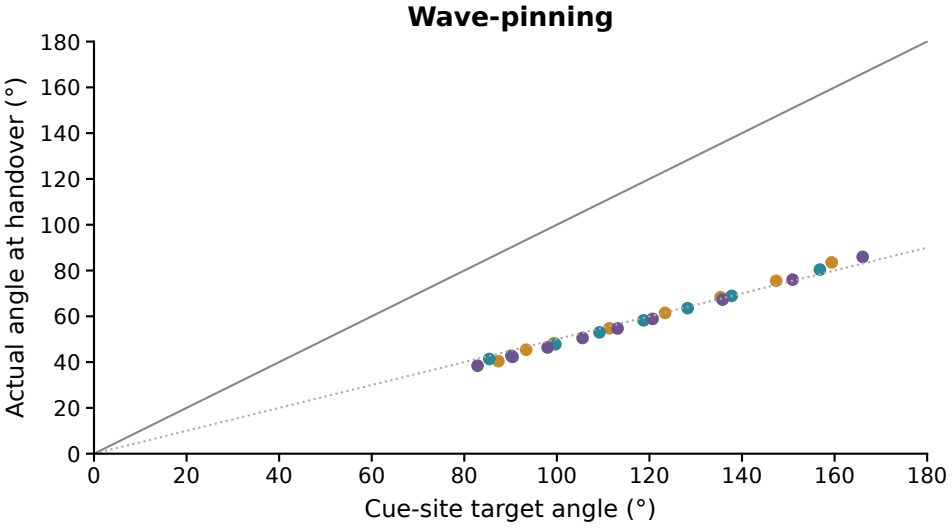
Primary gap: current global10%-contrast contours. ▲: resolution-time lower bound at completed horizon.
Alternate, merged and unseeded cases remain in the complete evidence table and native-history reader.

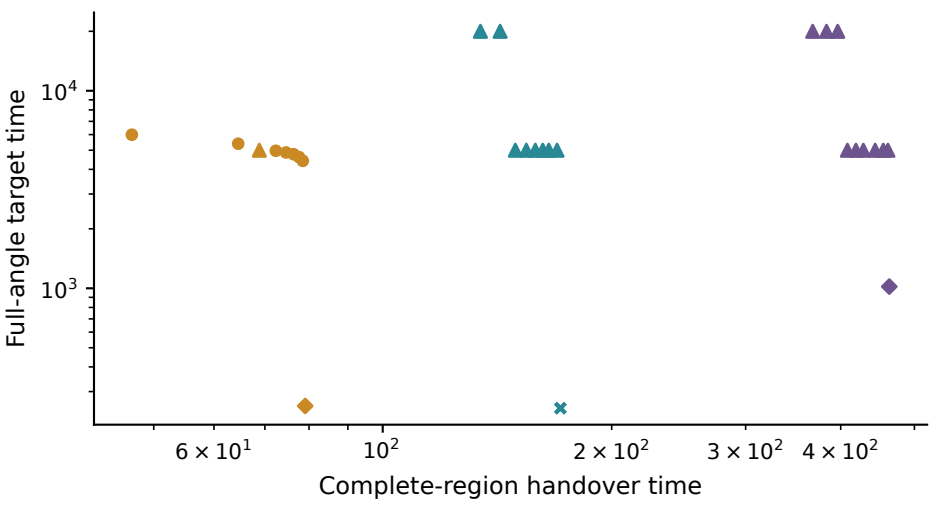

S3 · Fixed-angle handover remaps in all four models
reference size
1.26× native size label
1.59× native size label
Wave-pinning
Goryachev
Holmes 3
Spring
Handover time
Requested cue-site angle (°)
Actual angle / target
Solid: measured native size; dashed: reference-size handover curve evaluated at the larger cell's measured-width gap.
No extrapolation or interpolation through unavailable/censored neighbors. Insets: actual direction at handover / requested cue angle.

S4 · Handover typically reaches part of the requested angle; full-target timing is separate

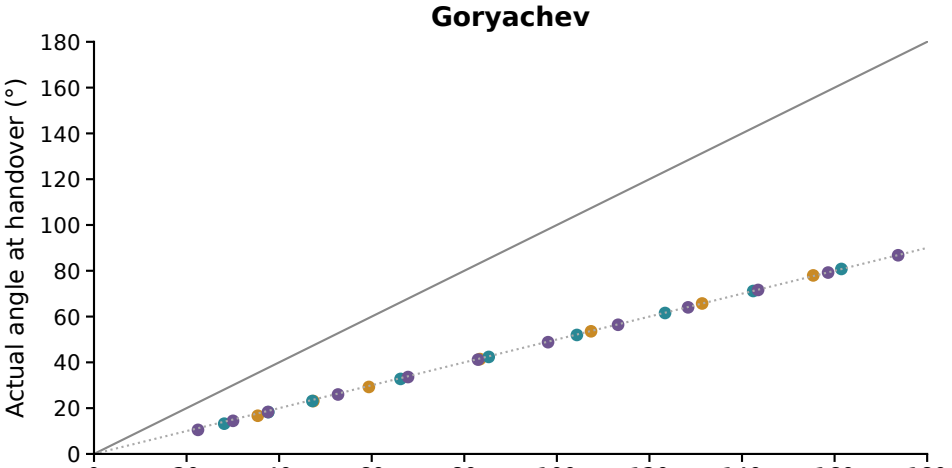
reference size · 1.26× native size label · 1.59× native size label

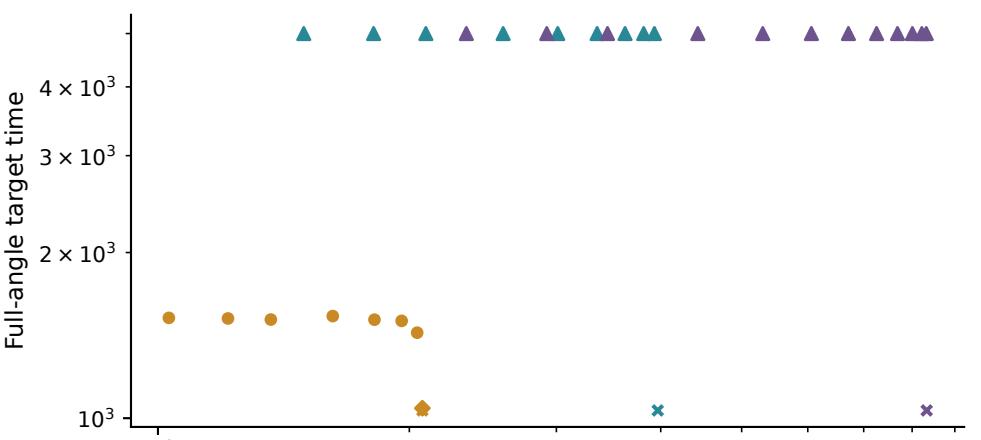
Wave-pinning



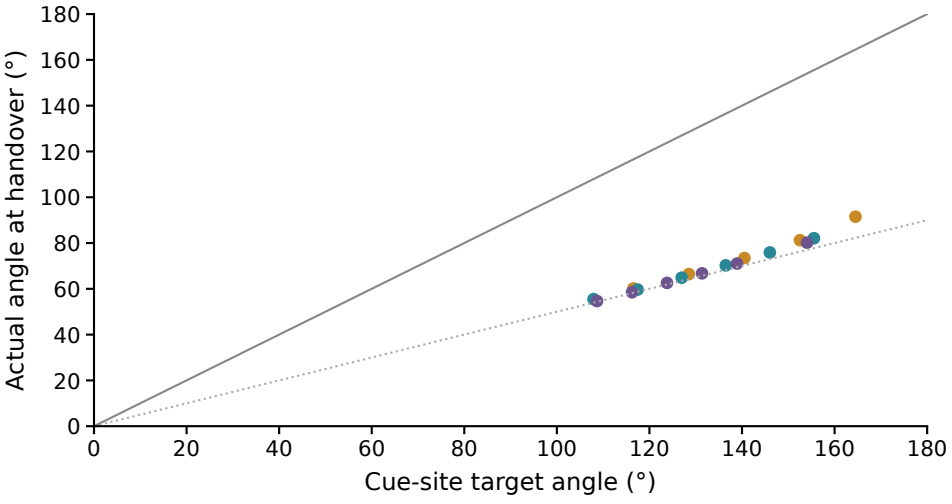


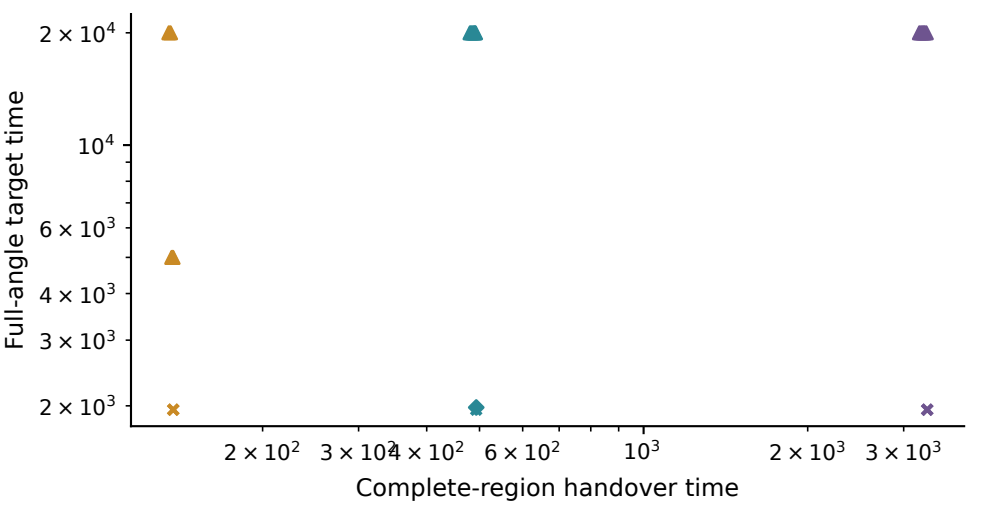
Goryachev



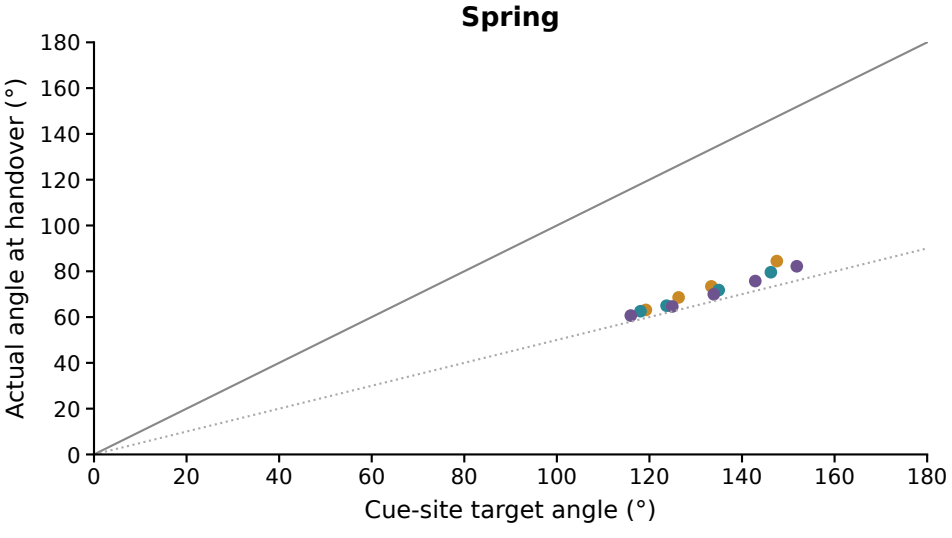


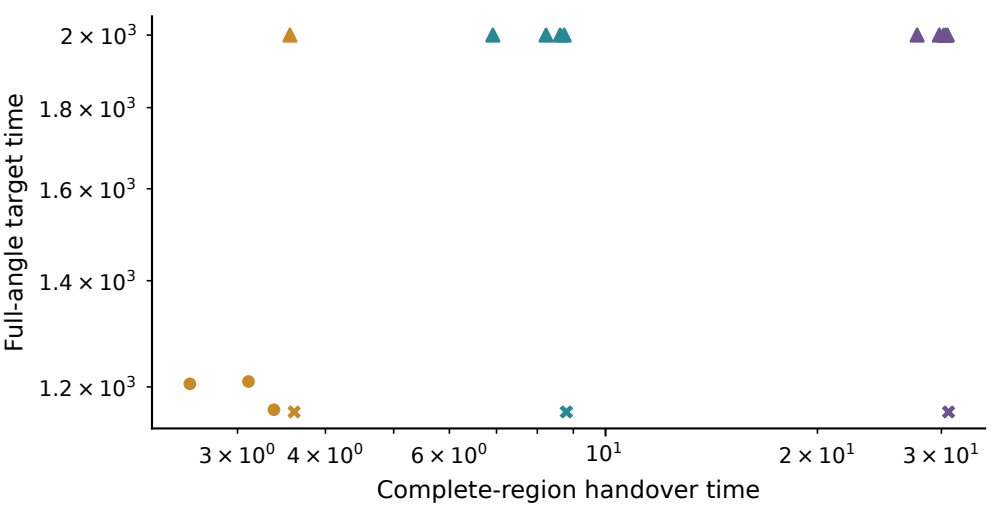
Holmes 3



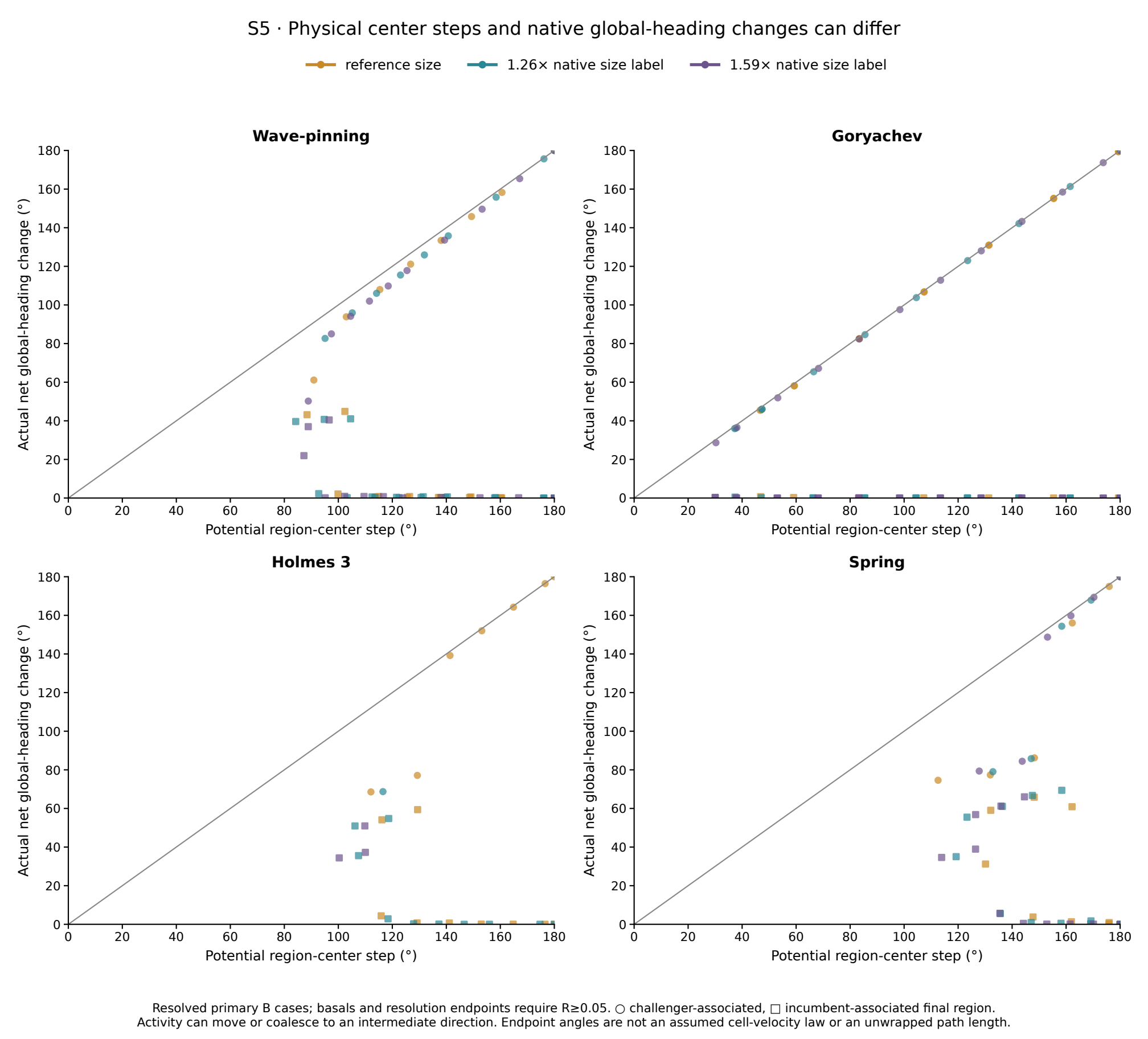


Spring

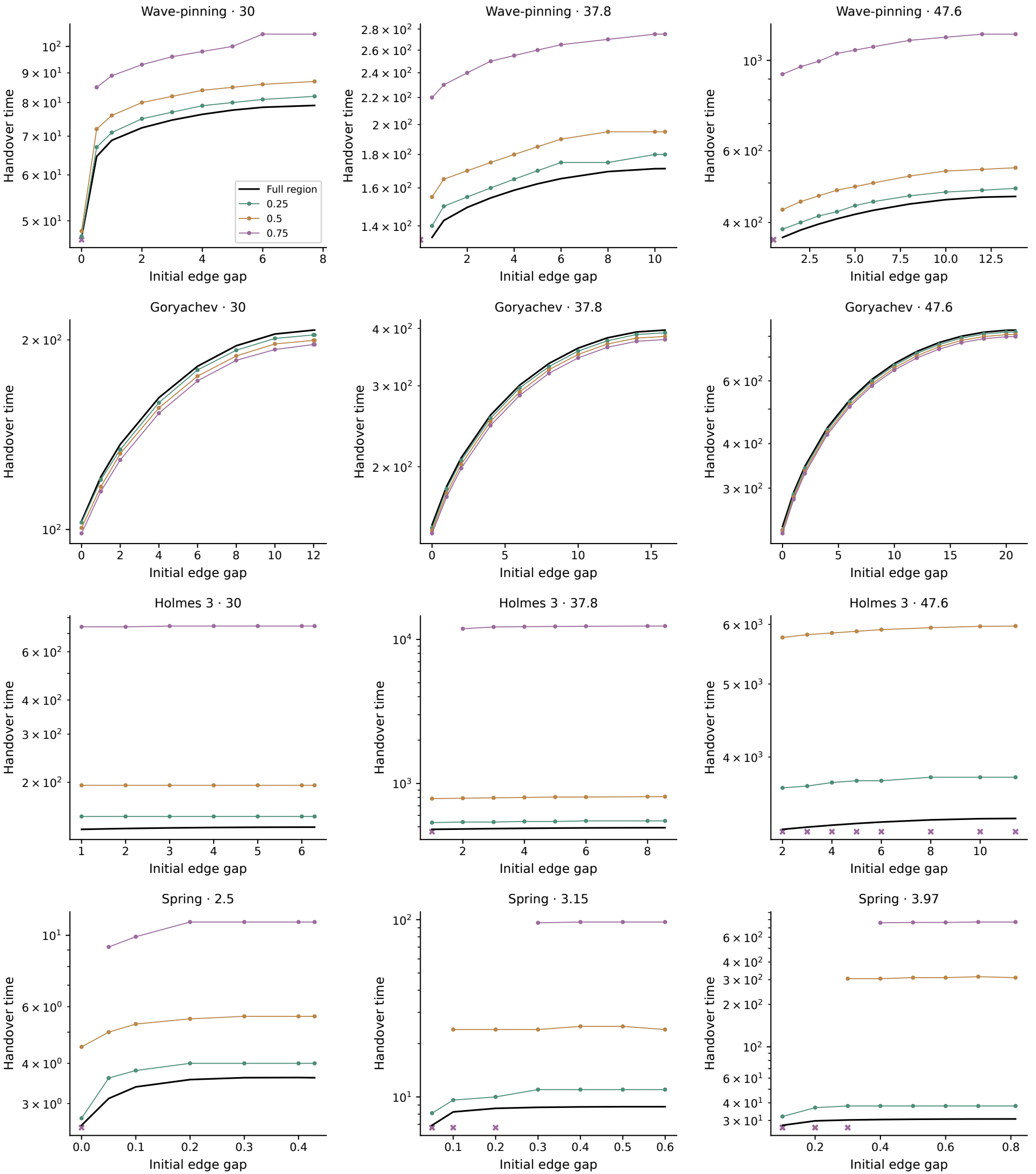




R=|M₁|/M₀≥0.05 at usable samples and intervening conservative complex-moment chords. ▲: finite-cap timing lower bound.
×: direction-quality/cancellation gap prevents a full timing bound. Exact-direction and0.1/0.2 gates plus50/90/99% targets are tabulated.

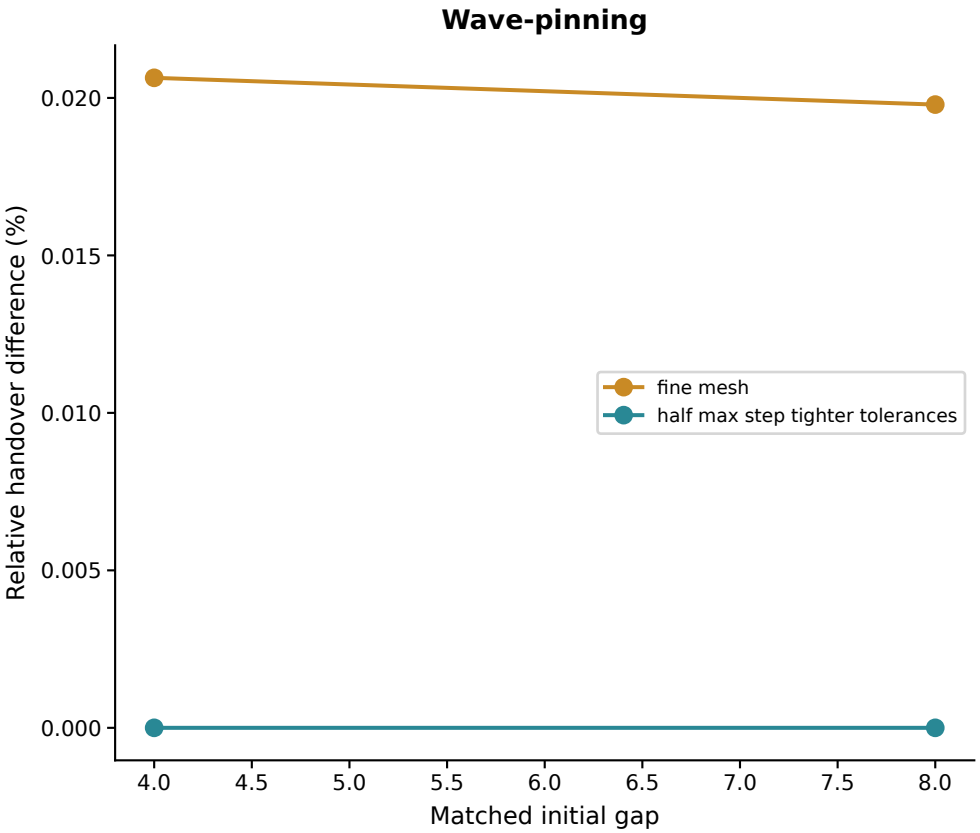

S5 · Physical center steps and native global-heading changes can differ
reference size
1.26× native size label
1.59× native size label
Wave-pinning
Goryachev
Holmes 3
Spring
Actual net global-heading change (°)
Potential region-center step (°)
0
20
40
60
80
100
120
140
160
180
Resolved primary B cases; basals and resolution endpoints require R≥0.05. ○ challenger-associated, □ incumbent-associated final region.
Activity can move or coalesce to an intermediate direction. Endpoint angles are not an assumed cell-velocity law or an unwrapped path length.

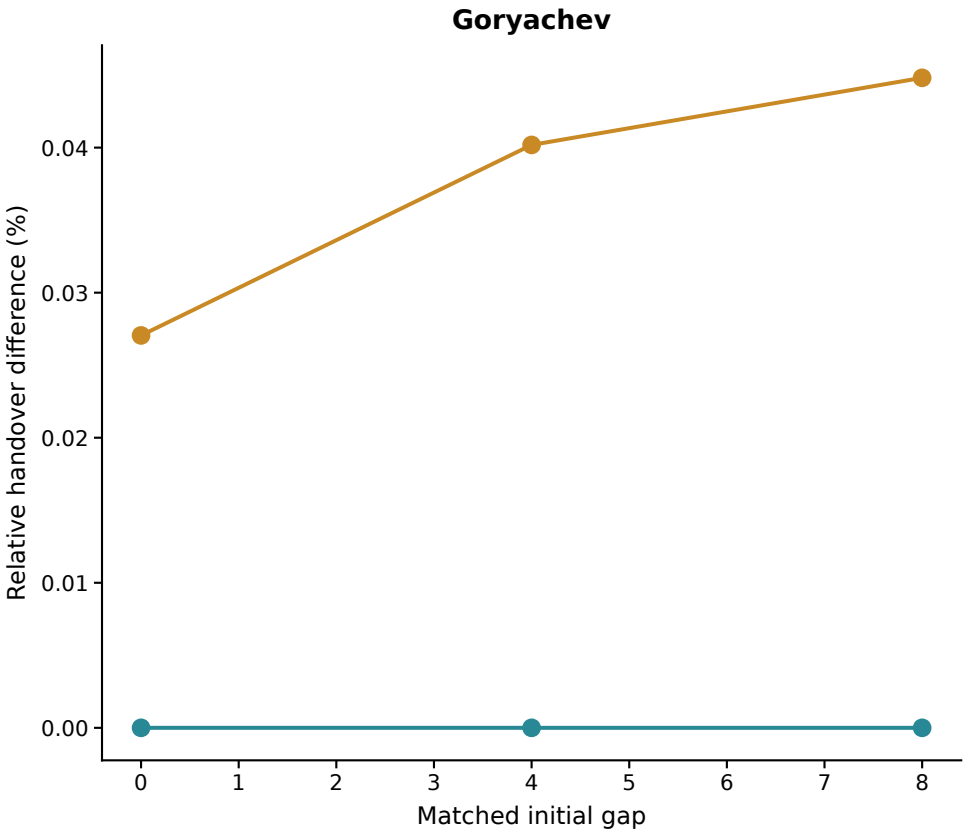
S6 · Amount sensitivity: local 25/50/75% peak contours versus complete-region excess



Original excess above each region's local-peak contour; watershed basins and common current floor are unchanged.
Times are retained native-sample upper brackets. ×: unavailable continuous sampled two-region crossing; no cap substitution.

S7 · Numerical, preparation and complete-source checks





Numerical evidence

Fresh native basals: 12 / 12 qualified
Long no-cue controls: 53,000 chemical / 5,300 Spring
Fine-mesh basals: 2 / 2 independently qualified
A1 refinement controls: 12 / 12 native audits
Observed control roots: 10 / 10 verified
A2 native audit: 545 / 545
B native audit: 381 / 381 (24 mirrors agree)
B mesh/step controls: 36 / 36 native audits

Readout and source gates

Native positivity and conserved-pool gates pass.
Scalar fate and field stationarity remain separate.
Continuous roots replay the original upper fields.
B controls retain all seed statuses and winners.
35 / 36 resolution brackets overlap.
Goryachev g=4, ratio 1, fine mesh: 1040–1060
versus 1080–1100 (about 4% timing sensitivity).
Full native atlas: 1,150 histories; all sources linked.